CLICK ENGINEERING

ACN 162 735 917
FIRST FLOOR, 25-27 RAILWAY ROAD, BLACKBURN, VIC. 3130
PHONE: (03) 9894 4222 FAX: (03) 9894 4424
Email: contact@clickeng.com.au

SITE PREPARATION NOTES

1. Top soil containing grass roots or other organic material shall be removed from the area on which the waffle raft is to rest.

2. Cut and fill site to form a level bench as per architectural drawings.

3. Where shallow surface fill exists, or is to be placed to level the site the following conditions must be satisfied:
   - **3.1.** **Sand Fill**. Up to 600mm of sand fill only may exist or be placed below the waffle raft provided that it has been well compacted in 150mm layers by a vibrating roller.
   - **3.2.** **Non Sand Fill.** Up to 300mm of non sand fill may exist or be placed below the waffle raft provided that it has been well compacted in 150mm layers by a static roller.
   - **3.3.** **Controlled Fill**. Controlled fill shall be placed in accordance with AS. 3798 following specific engineering advice.
   - 3.4. Where the above conditions cannot be complied with, the waffle raft will require to be supported on piers. This office should be contacted for further information regarding the size and location of piers.

4. The construction process may be made easier if the fill material extends one metre outside the building line. The area within one metre of the slab edge shall be graded away from the slab to prevent water resting against the slab.

GENERAL NOTES

1. Do not scale these drawings. All dimensions to be taken from architectural drawing.
2. The owner's attention is drawn to appendix B of AS. 2870 (current edition) `performance requirements and foundation maintenance'.
3. Refer to soil report, architectural drawings and detail sheets for additional construction and maintenance requirements which form part of this design.
4. Whilst every effort has been expended to locate and design for services in and around the proposed development, it remains the contractors responsibilty to ensure no services are damaged or exposed during the proposed works.

WAFFLE NOTES

1. Plumber to lay waste pipes below ground surface at minimum grades. Risers to be staked firmly. If plumbing risers are located within ribs then maintain minimum rib width of 110mm by cutting ends off void formers and lapping reinforcement as required. Void formers to be cut and taped around plumbing risers.

2. External service trenches are to be kept away from the house by a distance of at least the depth of the trench.

3. Place up to 50mm of quarry sand or quarry product over the building area within the edge and any internal load bearing beams.

4. Vapour barrier to be placed under void formers and ribs. Prepare waffle raft within formwork in accordance with footing plan and detail sheets.

5. Vapour Barrier shall be 0.2mm thick polyethylene sheeting with adequate impermeability, and durability to ultraviolet deterioration and impact during construction. Lapping shall be not less than 200mm at joints and penetrations by pipes or plumbing shall be taped. In accordance with AS2870 C5.3.3, the barrier is against vapour, not water, minor punctures are not important.

6. Prepare waffle void formers within formwork boards in accordance with footing plan and detail sheets. Void formers to be spaced using appropriate plastic spacers.

7. Use appropriate plastic based bar chairs to ensure 20mm minimum top cover to slab mesh. Slab fabric shall be lapped by one full panel of fabric in both directions and laps staggered to avoid a build up of mesh. Trench mesh in beams shall be placed with 40mm cover top and bottom and spliced, where necessary, by a lap of 500mm. At T- and L- intersections the trench mesh shall be overlapped by the width of the fabric.

8. Concrete shall be not less than 20 MPa grade (unless noted otherwise) with 20mm nominal maximum aggregate size and 100mm slump. Concrete should be cured for at least 7 days prior to placing brickwork.

9. Refer to sections 5 and 6 of AS. 2870-(current edition) for full set of detailing and construction requirements.

10. Shrinkage cracking to any concrete surface can occur as part of the natural curing process. This is not considered a structural defect and can be expected to occur within the guidelines specified in the Residential Slabs and Footing Code (AS2870), in particular Appendix C, Table C2.

DRAINAGE REQUIREMENTS

1. Surface drainage of the site shall be controlled from the start of site preparation and construction. The drainage system shall be completed by the finish of construction of the building.

2. Surface drainage shall be constructed to avoid water ponding against or near the footing. The ground in the immediate vicinity of the perimeter footing, including the ground uphill from the slab on cut and fill sites, shall be graded to fall 50mm away from the footing over a distance of 1m and shaped to prevent ponding of water. Any perimeter paving shall also be suitably sloped away from the footings.

3. Where perimeter paving slabs are constructed they must have a 10mm layer of ableflex or similar between the paving slab and the footing/wall. An agricultural drain or similar must be installed at the outer edge of the perimeter of the paving slab and is to be diverted to the legal point of discharge.

4. Watering and garden beds are not permitted adjacent to building(s) and adjacent to perimeter paving slabs. Subsurface drains shall not be used within 1.5m of the building unless designed by a qualified engineer familiar with this type of design.

5. Bulk excavations incorporating cut and fill must have an agricultural drain (or swale drain) at the base of the cut into the natural clay and all surface and sub-surface water is to be diverted from the footing to the legal point of discharge. Provision of a dish drain to the top of all batters will prevent the occurrence of scouring on the batter faces.

6. The base of trenches shall be sloped away from the building. Trenches shall be backfilled with clay in the top 300mm within 1.5m of the building. Compacted clay should be used for backfilling where pipes pass beneath the footing system the trenches will be backfilled full depth with clay or concrete to restrict the ingress of water beneath the footing system.

7. Penetrations of the edge beams of raft or perimeter strip footings shall be avoided where practicable, but where necessary shall be detailed to allow for movement.

8. Drains attached to/or emerging from underneath the building shall incorporate flexible joints immediately outside the footing and commencing within one metre of the building perimeter to accommodate a total range of differential movement in any direction equal to the estimated characteristic surface movement of the site (ys). In the absence of a specific design guidance the fittings or other devices that are provided to allow for movement shall be set at the mid position of their range of possible movement at the time of installation, so as to allow for movement equal to 0.5ys in any direction from the initial setting. This requirement applies to all stormwater, sewer drains and discharge pipes.

LOT 31, OSKAR COURT, PAKENHAM

SIMONDS HOMES EAST MELBOURNE

Revision	Date	Description
2214421/2	13/09/21	Revision
2214421/1	03/09/21	Revision
2214421	25/08/21	Construction Issue

Drawn
SHAUN .F
Design
MANA .M
Date
13/09/21
Scale ---

Title Sheet

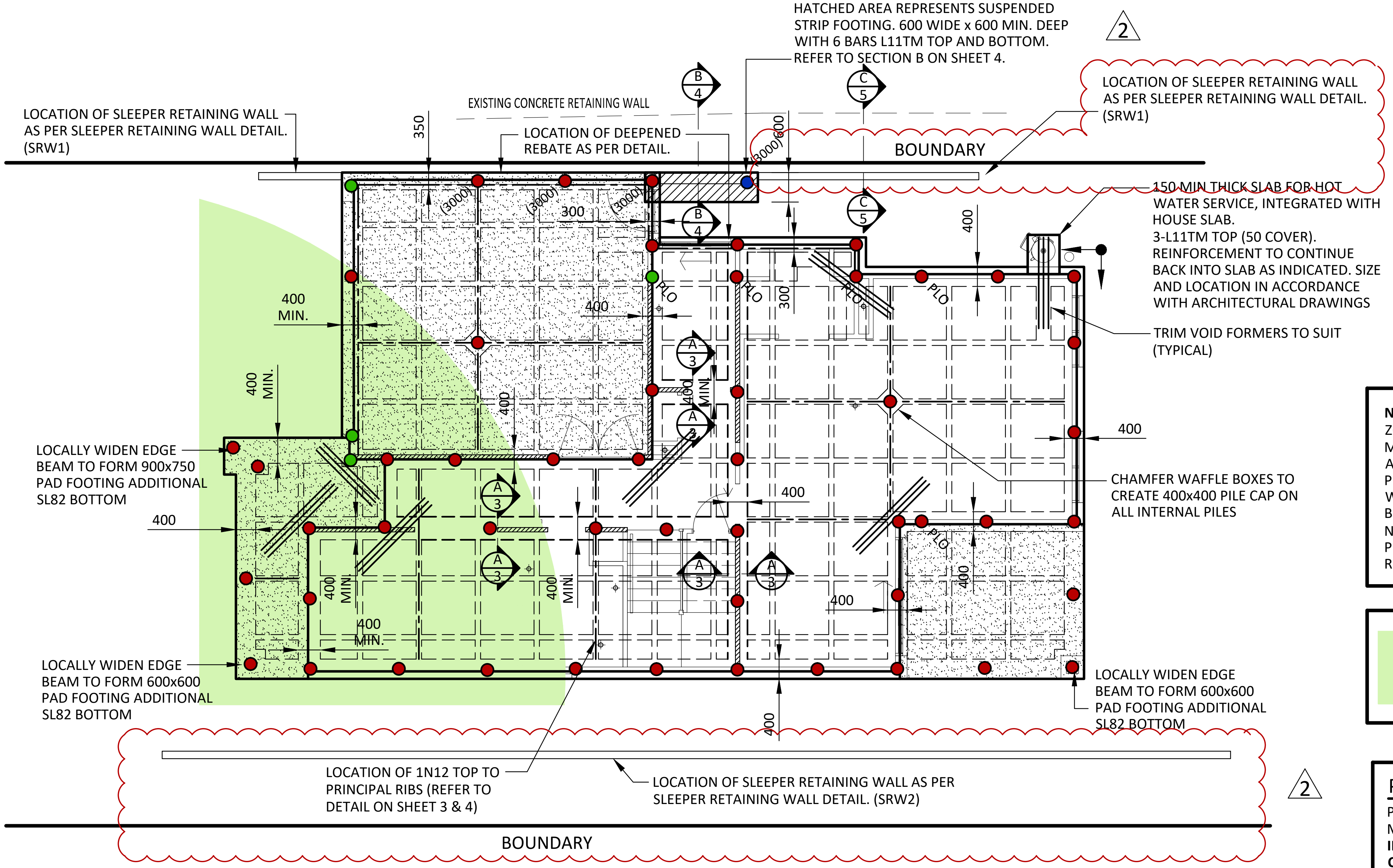
Checked by, for and on behalf of Click Engineering (aust) Pty Ltd

Sheet No. **1**

Job No: **2214421/2**
SIMONDS HOMES EAST MELBOURNE
LOT 31, OSKAR COURT, PAKENHAM

SITE CLASSIFICATION	CLASS P (MOD. M)
FOOTING SYSTEM	SUSPENDED WAFFLE RAFT
CLASSIFIER	MACGREGOR GEO P/L
SOIL REPORT NUMBER	2213991

MM01-dwg to pdf.pc3



SCREW IN PILE SCHEDULE

- RED SCREW-IN PILE AS PER MANUFACTURERS SPECIFICATIONS, TAKEN DOWN THROUGH ANY EXISTING FILL AND DRIVEN TO A SAFE WORKING LOAD OF **80kN**.
  BUT ALSO BELOW THE ANGLE OF REPOSE OF ADJOINING RETAINING WALL MIN. DEPTHS SHOWN IN BRACKETS.
- GREEN SCREW-IN PILE AS PER MANUFACTURERS SPECIFICATIONS, TAKEN DOWN THROUGH ANY EXISTING FILL AND DRIVEN TO A SAFE WORKING LOAD OF **100kN**.
- BLUE SCREW-IN PILE AS PER MANUFACTURERS SPECIFICATIONS, TAKEN DOWN THROUGH ANY EXISTING FILL AND DRIVEN TO A SAFE WORKING LOAD OF **70kN**.
  BUT ALSO BELOW THE ANGLE OF REPOSE OF ADJOINING RETAINING WALL MIN. DEPTHS SHOWN IN BRACKETS.

NOTE: WHERE EXISTING OR PROPOSED TREES ARE WITHIN THE ZONE OF INFLUENCE OF ANY FOOTINGS (i.e. WITHIN 1.0xTHEIR MATURE HEIGHT), THEN THE FOOTINGS ARE TO BE DEEPENED AND FOUNDED DIRECTLY ONTO WEATHERED BEDROCK IF PRESENT OR 2000mm BELOW NATURAL GROUND LINE, WHICHEVER IS SHALLOWER. ALTERNATIVELY THESE TREES COULD BE REMOVED OR TREE ROOT BARRIERS PLACED.
N.B. ARBORIST ADVICE AND/OR COUNCIL PERMIT IS REQUIRED PRIOR TO PLACEMENT OF TREE ROOT BARRIERS, CUTTING TREE ROOTS, OR REMOVING TREES.

NOTE:
ALL SCREW IN PILES LOCATED WITHIN SHADED AREAS ARE TO BE FOUNDED AS INDICATED BUT ALSO AT LEAST **2000mm** MIN BELOW THE NATURAL GROUND LINE OR DIRECTLY ONTO WEATHERED ROCK DUE TO EXISTING OR PROPOSED TREES.

2

POINT LOADS:-

POINT LOADS FOUNDED DIRECTLY ONTO WAFFLE SLAB PANELS MUST NOT EXCEED 10kN SAFE WORKING LOAD.
IF SO, THIS OFFICE SHOULD BE CONTACTED FOR FURTHER CONSULTING.

NOTE:
CONSULTATION WITH THE GEOTECHNICAL ENGINEER HAS BEEN SOUGHT TO ENSURE THAT THE UNDERLYING SITE CLASSIFICATION IS NOT AFFECTED BY THE PROPOSED SITE WORKS. THE PROPOSED SITE WORKS HAVE BEEN TAKEN INTO ACCOUNT AT THE TIME OF THIS FOOTING DESIGN.

NOTE:
AT A RE-ENTRANT CORNER WHERE AN EXTERNAL BEAM CONTINUES AS AN INTERNAL BEAM, THE EXTERNAL BEAM DETAILS SHALL BE CONTINUED FOR A LENGTH OF 1m INTO THE INTERNAL BEAM.

⌖ INDICATIVE LOCATION OF WASTEPOINT, REFER TO ARCHITECTURAL DRAWINGS FOR EXACT LOCATION

FILL NOTE:
ALL LOOSE SURFACE FILL, ALL ROOTS AND ALL ORGANIC MATERIAL ARE TO BE REMOVED FROM THE BUILDING PLATFORM. SITE IS TO BE CUT AND/OR FILLED (OR SITE SCRAPED) TO FORM A LEVEL BENCH. ANY EXISTING AND ANY PLACED FILL MATERIAL IS TO CONFORM AND BE PLACED IN ACCORDANCE WITH AS2870-2011 CL.6.4.2 AND IS NOT TO EXCEED 300mm. ANY IMPORTED PLACED FILL IS TO BE A WELL GRADED COMPACTIBLE MATERIAL. PLACED FILL IS TO CONTINUE PAST THE EDGE OF THE BUILDING BY AT LEAST 1m AND IS TO BE PLACED IN ACCORDANCE WITH AS2870-2011 CL.6.4.2. IF THE BUILDING PLATFORM IS SOFT OR THE TOTAL DEPTH OF FILL IS GREATER THAN 300mm THEN THIS OFFICE IS TO BE CONTACTED FOR REDESIGN.

IN ACCORDANCE WITH NCC VOL.2 SECTION 3.1. SITE BATTERS MUST NOT EXCEED:
- SAND AND SILT, 1:2
- STIFF CLAY, 1:1 FOR CUT - 1:2 FOR FILL BATTER

IF THIS CANNOT BE ACHIEVED ON SITE THEN RETAINING WALLS WILL BE REQUIRED AND THIS OFFICE IS TO BE CONTACTED FOR FURTHER CONSULTATION

WAFFLE PARAMETERS

SLAB THICKNESS	- 85mm
BOX HEIGHT	- 225mm
OVERALL DEPTH	- 310mm
MAX. RIB CENTRES	- 1200mm
INTERNAL RIB WIDTH	- 110mm

REINFORCEMENT

SLAB MESH	- SL82
INTERNAL RIB	- 1-N12 BOTTOM
EXTERNAL RIB	- Refer to details for width and reinforcement

ADDITIONAL REINFORCEMENT REQUIREMENTS FOR EDGE BEAMS WITH WIDTHS EXCEEDING 300mm

STEM WIDTH mm	ADDITIONAL TOP STEEL	BEAM BASE WIDTH mm	ADDITIONAL BOTTOM STEEL
110-150mm	-	300-330mm	-
151-220mm	1-N12	331-440mm	1-N12
221-330mm	2-N12	441-550mm	2-N12
331-440mm	3-N12	551-660mm	3-N12

LEGEND

- DENOTES STARTING POINT FOR LAYING OF VOID FORMERS.
- DENOTES 3-L11TM (OR 3-N12 BARS) x 2000mm LONG TIED TO UNDERSIDE OF SLAB MESH. BARS MAY BE DELETED WHERE SLAB MESH LAPS AT INTERNAL CORNERS.
- SHADED AREA DENOTES AREA WHERE VOID FORMERS ARE SET DOWN. REFER TO DETAILS.
- LBW — LOAD BEARING WALL
- INDICATIVE LOCATION OF CUT FILL LINE

Drawn
SHAUN .F
Design
MANA .M
Date
13/09/21
Scale 1:100

Slab and Footing Plan

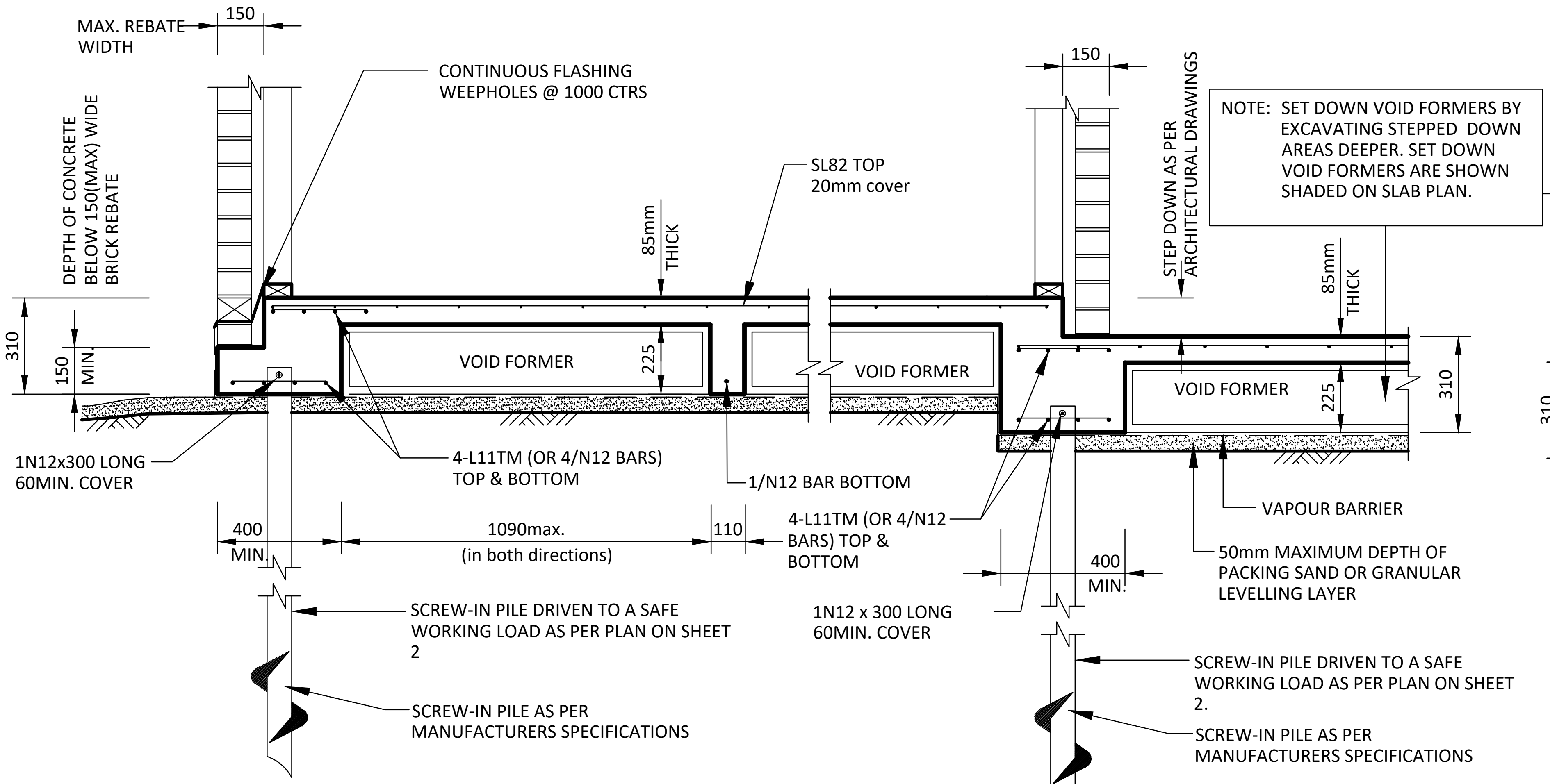
Sheet No. **2**

Job No: **2214421/2**

SIMONDS HOMES EAST MELBOURNE
LOT 31, OSKAR COURT, PAKENHAM

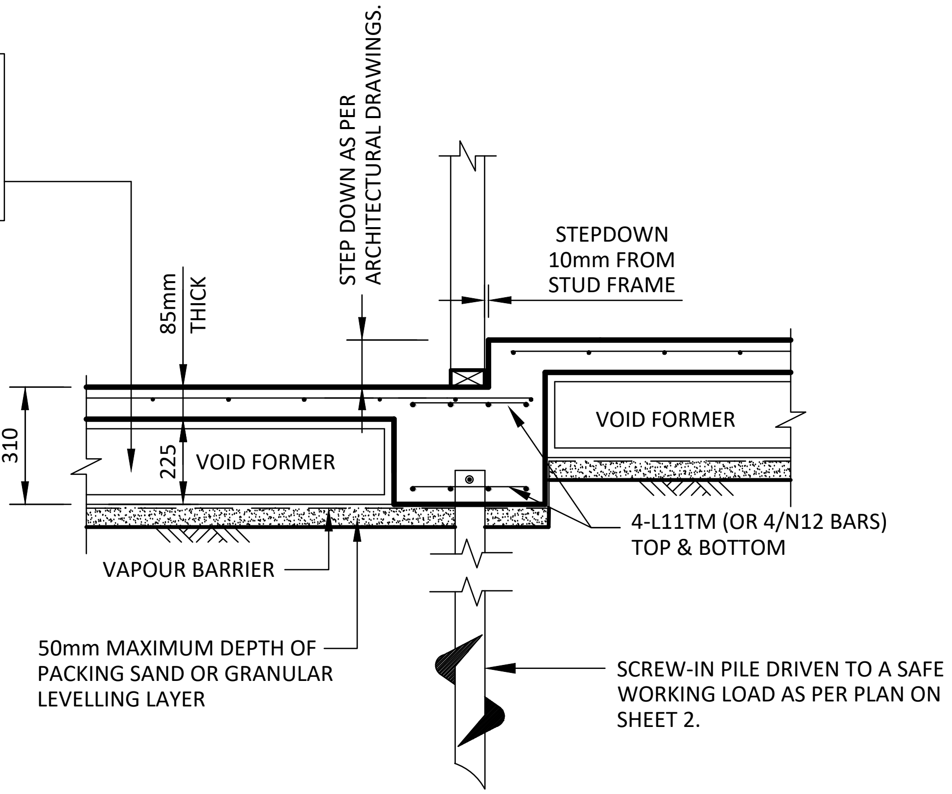
ACN 162 735 917
1st FLOOR, 25-27 RAILWAY RD,
BLACKBURN, VIC. 3130
E: contact@clickeng.com.au
P: (03) 9894 4222

CLICK ENGINEERING

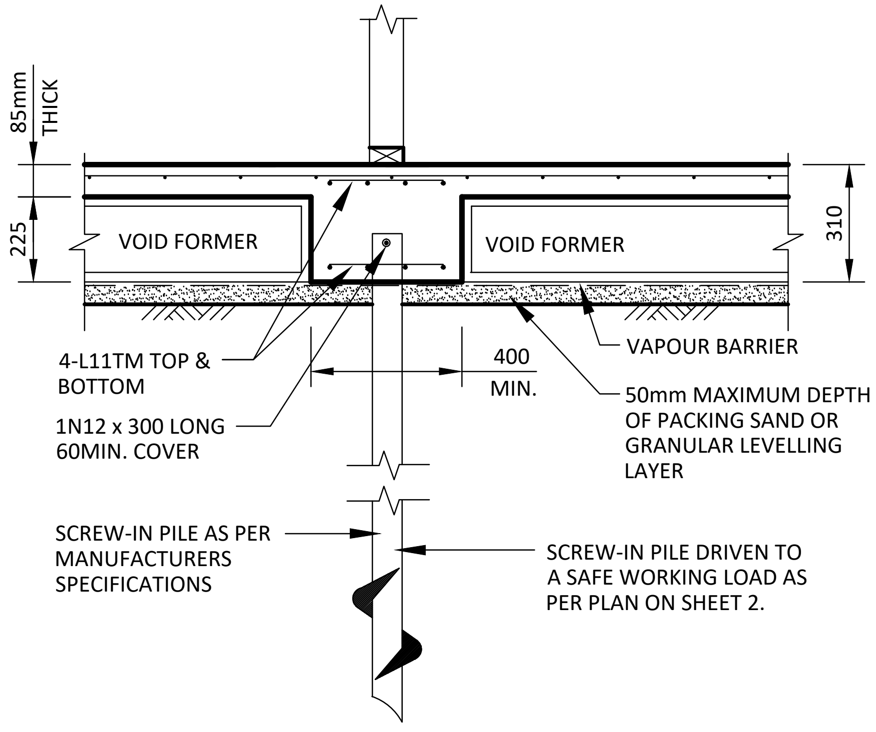


SUSPENDED WAFFLE SLAB CROSS SECTION

(DOUBLE STOREY SECTION)

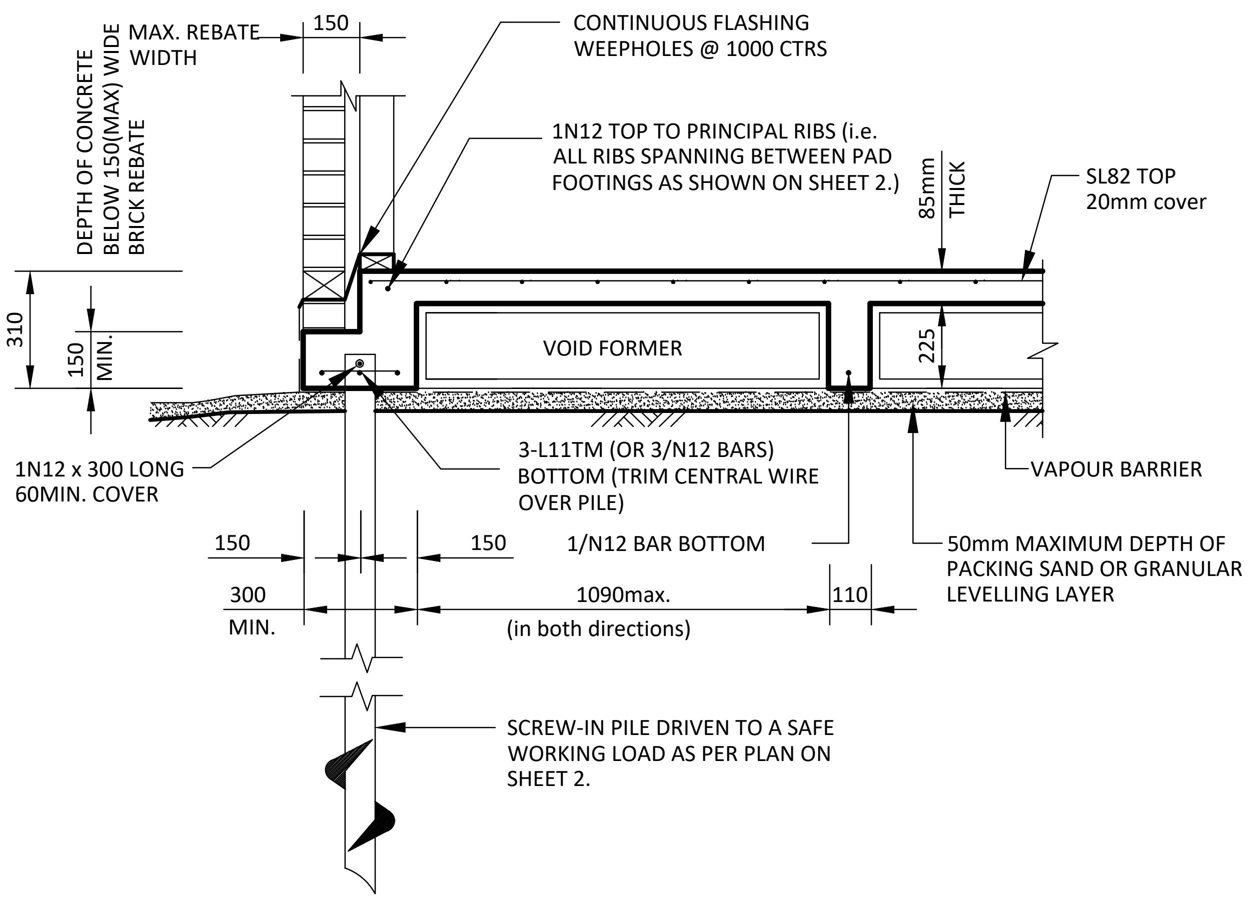


GARAGE STEP DOWN DETAIL SCALE 1:20



SECTION A/2

SCALE 1:20



SUSPENDED WAFFLE SLAB CROSS SECTION

(SINGLE STOREY SECTION)

Drawn SHAUN .F
Design MANA .M
Date 13/09/21
Scale 1:20

Associated Details 1

Sheet No. **3**

Job No: **2214421/2**

SIMONDS HOMES EAST MELBOURNE

LOT 31, OSKAR COURT, PAKENHAM

ACN 162 735 917
1st FLOOR, 25-27 RAILWAY RD,
BLACKBURN, VIC. 3130
E: contact@clickeng.com.au
P: (03) 9894 4222

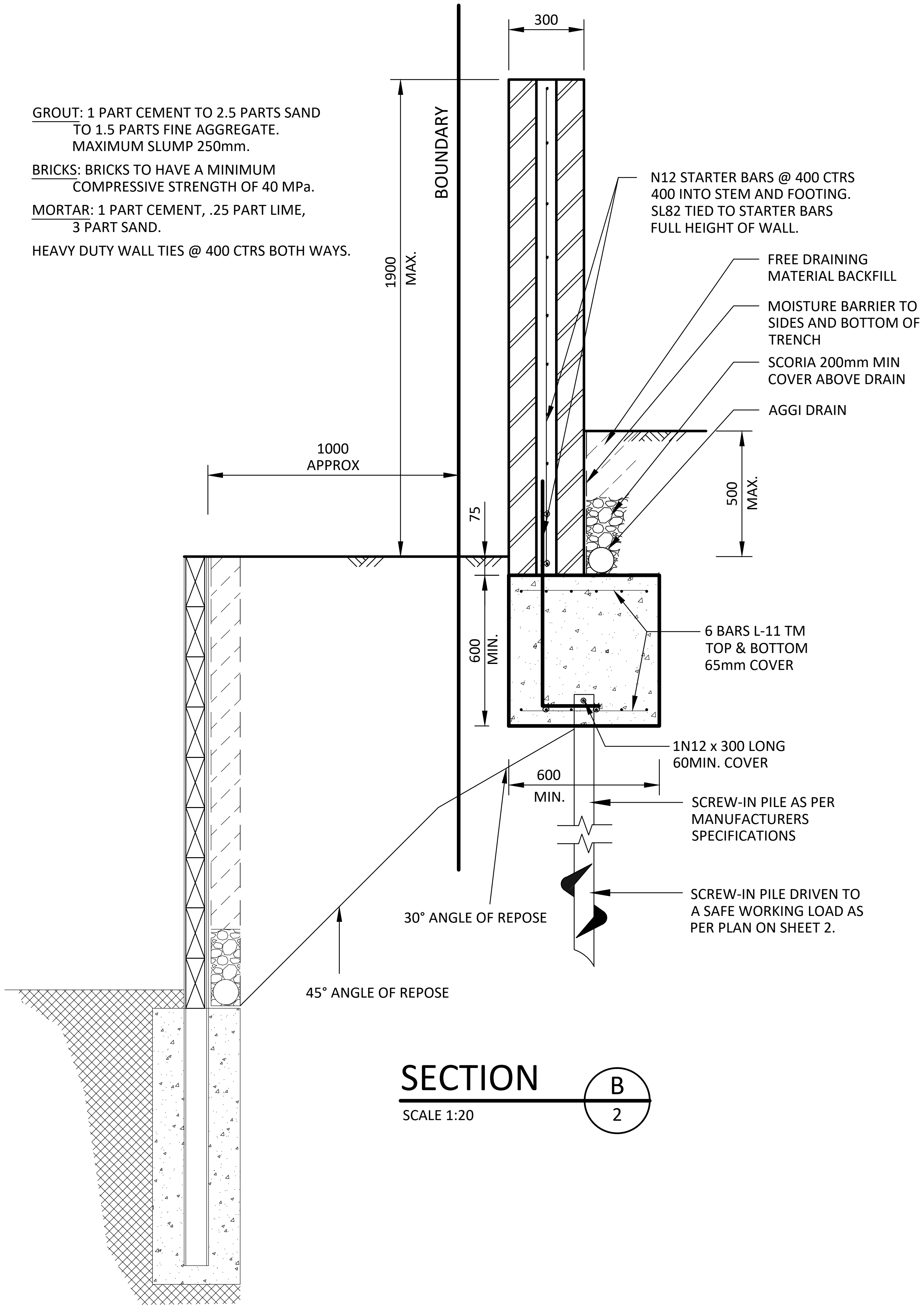
CLICK ENGINEERING

GROUT: 1 PART CEMENT TO 2.5 PARTS SAND TO 1.5 PARTS FINE AGGREGATE. MAXIMUM SLUMP 250mm.

BRICKS: BRICKS TO HAVE A MINIMUM COMPRESSIVE STRENGTH OF 40 MPa.

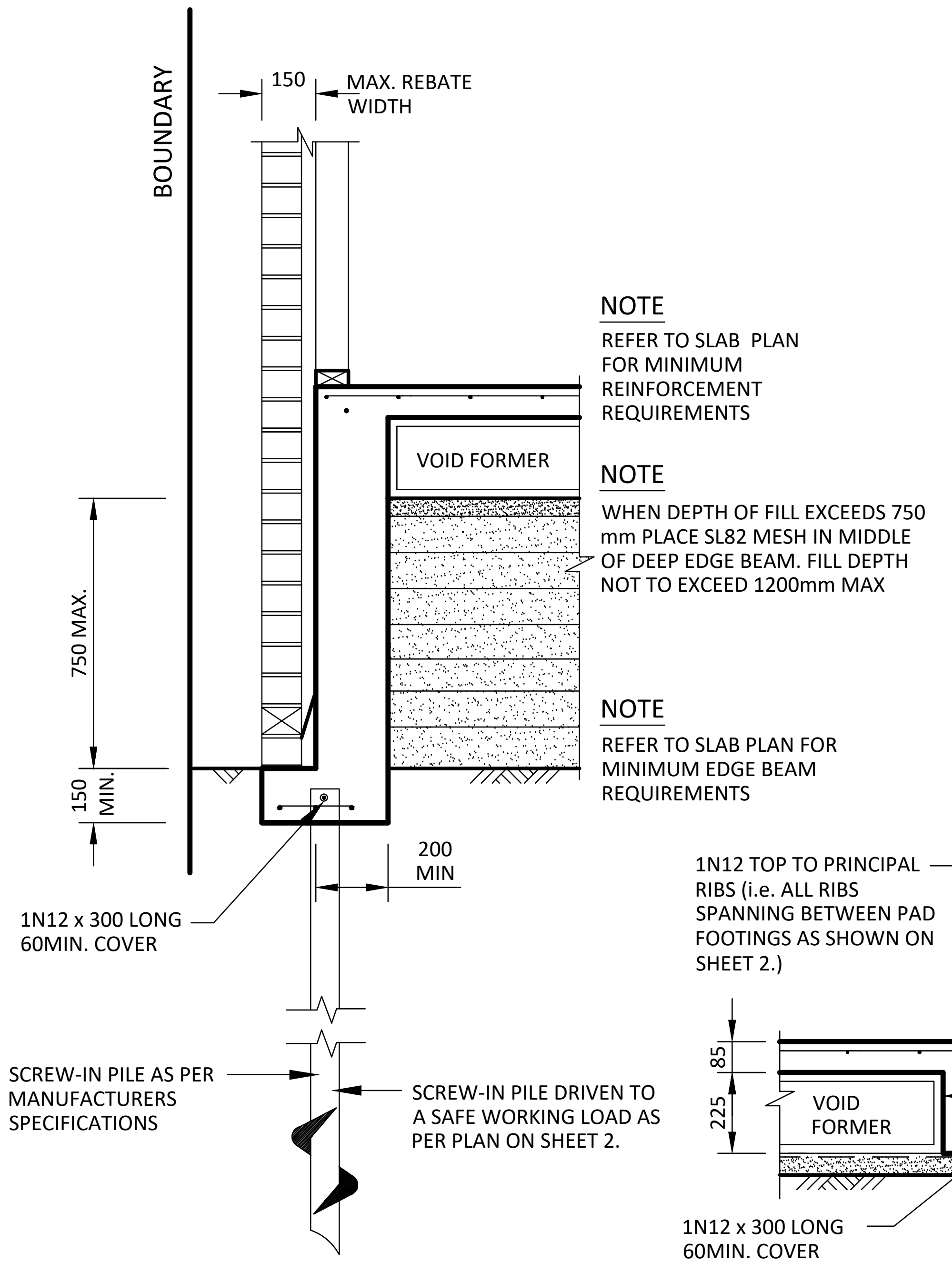
MORTAR: 1 PART CEMENT, .25 PART LIME, 3 PART SAND.

HEAVY DUTY WALL TIES @ 400 CTRS BOTH WAYS.



SECTION
SCALE 1:20

B
2



NOTE

REFER TO SLAB PLAN FOR MINIMUM REINFORCEMENT REQUIREMENTS

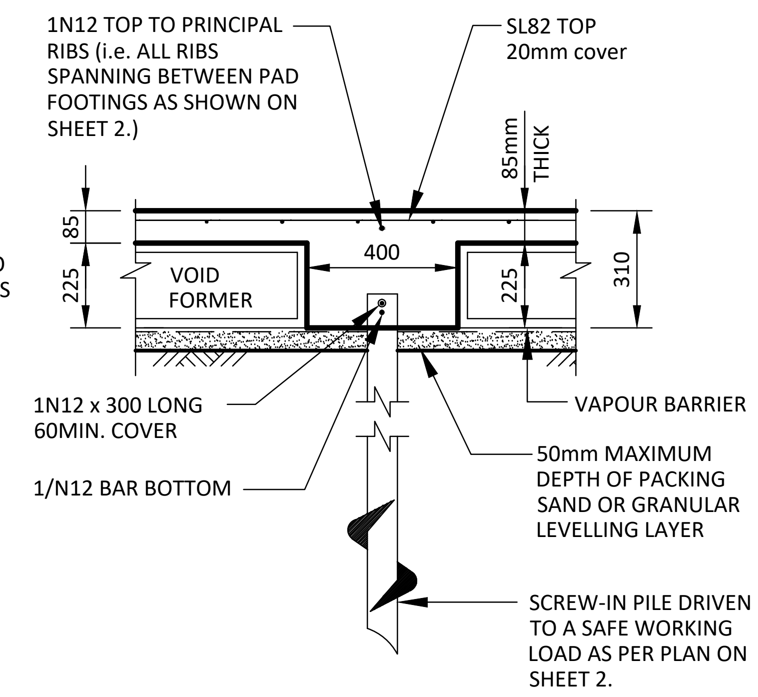
NOTE

WHEN DEPTH OF FILL EXCEEDS 750 mm PLACE SL82 MESH IN MIDDLE OF DEEP EDGE BEAM. FILL DEPTH NOT TO EXCEED 1200mm MAX

NOTE

REFER TO SLAB PLAN FOR MINIMUM EDGE BEAM REQUIREMENTS

DEEPENED REBATE DETAIL
(SINGLE STOREY SECTION)



TYPICAL PRINCIPAL RIB CROSS SECTION

Drawn SHAUN .F **Design** MANA .M **Date** 13/09/21 Scale 1:20	Associated Details 2	Sheet No. **4** Job No: **2214421/2** SIMONDS HOMES EAST MELBOURNE **LOT 31, OSKAR COURT, PAKENHAM**

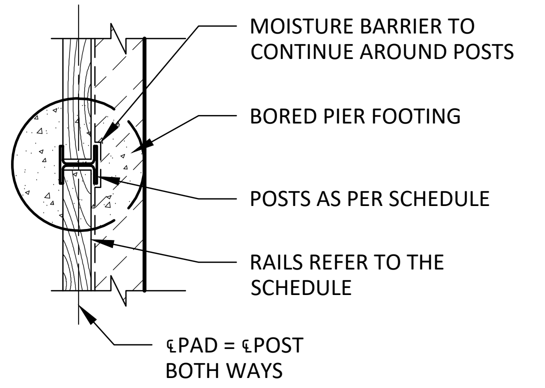
ACN 162 735 917
1st FLOOR, 25-27 RAILWAY RD, BLACKBURN, VIC. 3130
E: contact@clickeng.com.au
P: (03) 9894 4222

CLICK ENGINEERING

RETAINING WALL SCHEDULE (SRW1)

MAX. WALL HEIGHT	POST SIZE	POST CENTRES	RAILS	PAD SIZE	PAD DEPTH
600 mm	100 UC 15	1800 mm	200 x 75 F7	450 Ø	1000
800 mm	100 UC 15	1800 mm	200 x 75 F7	450 Ø	1200
1000 mm	150 UC 23	1800 mm	200 x 75 F7	450 Ø	1500

NOTE: IF THE SLEEPER RETAINING WALL SPANS ONLY 1 BAY AT 1800mm WIDE, 100UC15 POST MAY BE ADOPTED EACH END OF WALL AT 1000mm HIGH FOR A SINGLE BAY SPAN ONLY.



PLAN:- STEEL POST

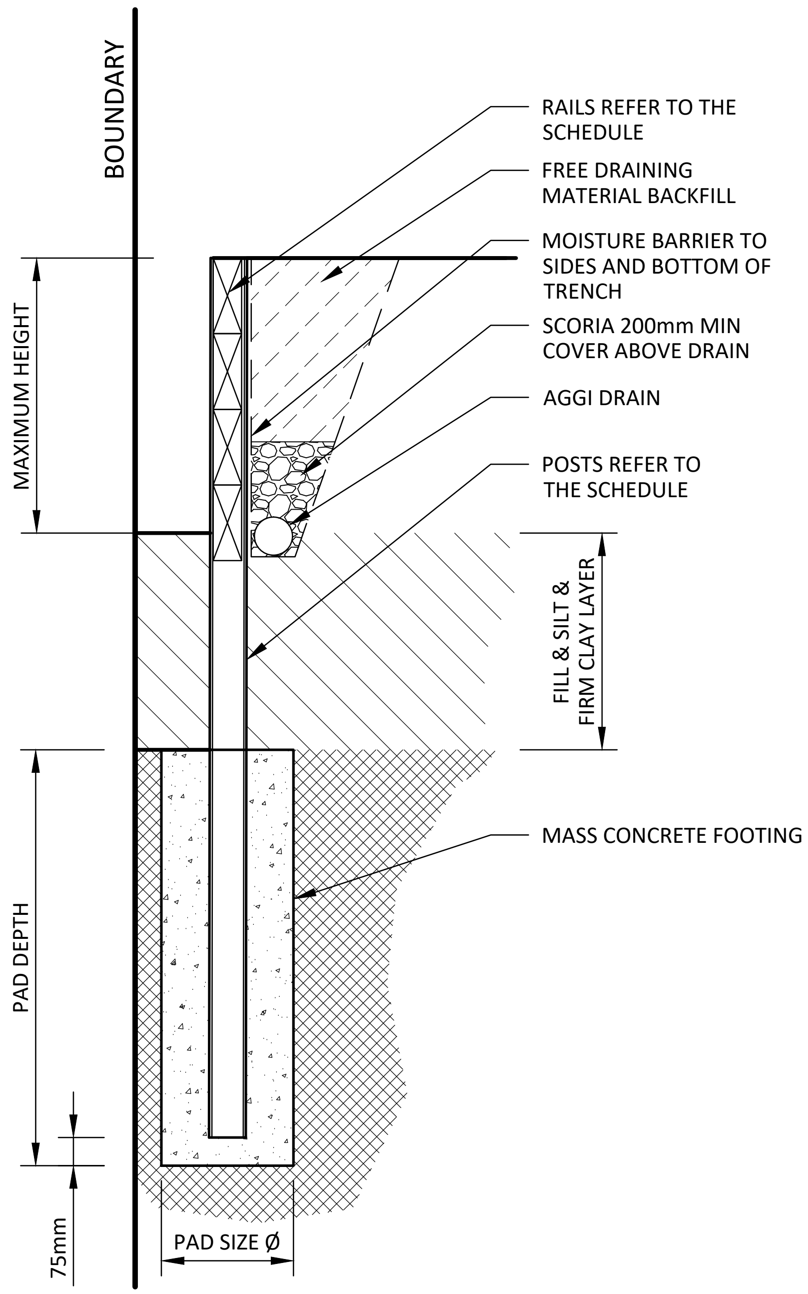
RETAINING WALL 1 NOTES:-

ALL POST PADS ARE TO BE FOUNDED FULL DEPTH AS NOTED IN THE SCHEDULE INTO THE **UNDERLYING NATURAL STIFF TO VERY STIFF CLAY. (200 KPa.)**

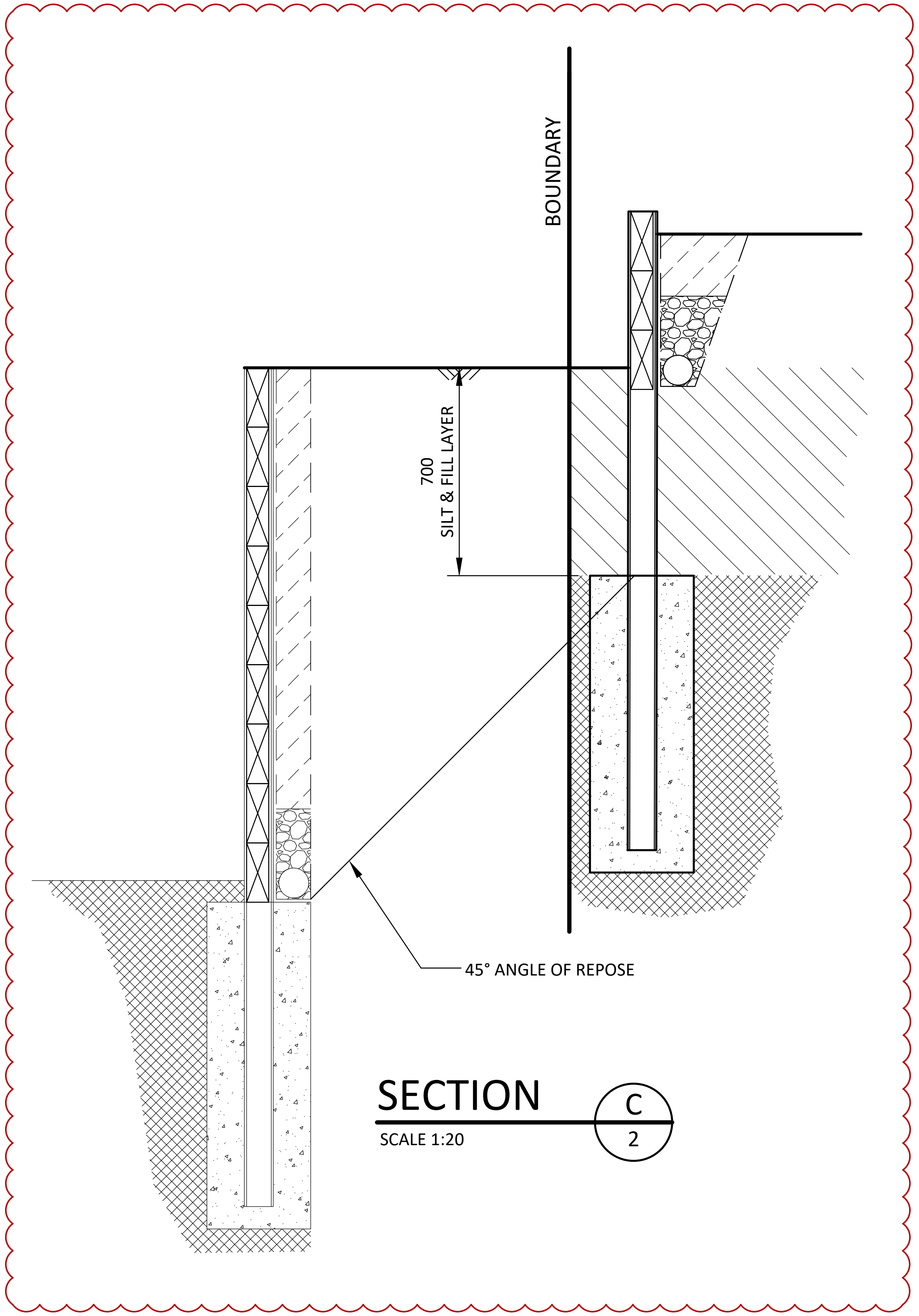
PADS AND A.G. DRAINS MUST NOT ENCROACH THE TITLE BOUNDARY.

RETAINING WALLS HAVE BEEN DESIGNED FOR A MAXIMUM SURCHARGE OF **5.0 KPa.**

FOR MAXIMUM RETAINING WALL DESIGN HEIGHT REFER TO SCHEDULE, SHOULD THE MAXIMUM HEIGHT BE EXCEEDED THE ENGINEER IS TO BE NOTIFIED FOR FURTHER ADVICE.



SLEEPER RETAINING WALL 1



SECTION

C
2

SCALE 1:20

2

Drawn SHAUN .F **Design** MANA .M **Date** 13/09/21 Scale ---	Retaining Wall Details	Sheet No. **5** Job No: **2214421/2** SIMONDS HOMES EAST MELBOURNE **LOT 31, OSKAR COURT, PAKENHAM**

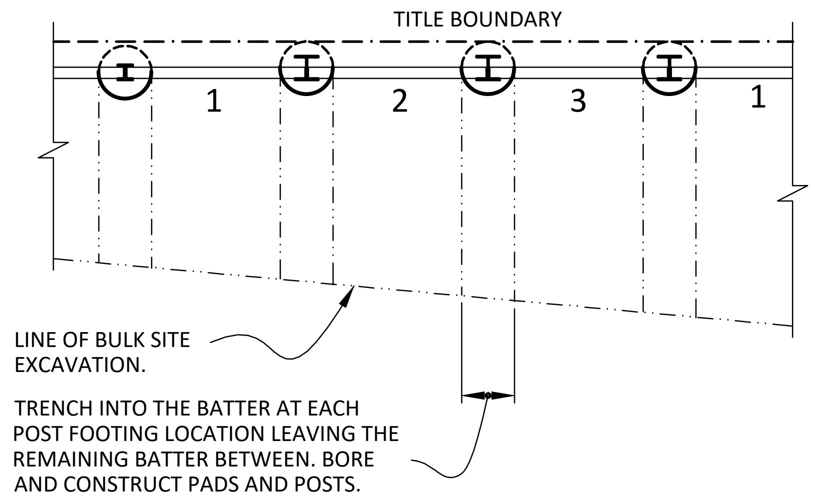
ACN 162 735 917
1st FLOOR, 25-27 RAILWAY RD,
BLACKBURN, VIC. 3130
E: contact@clickeng.com.au
P: (03) 9894 4222
CLICK ENGINEERING

RETAINING WALL SCHEDULE (SRW2)

MAX. WALL HEIGHT	POST SIZE	POST CENTRES	RAILS	PAD SIZE	PAD DEPTH
1000 mm	100 UC 15	1200 mm	200 x 75 F7	350 Ø	1500
1200 mm	100 UC 15	1200 mm	200 x 75 F7	350 Ø	1900
1400 mm	150 UC 23	1200 mm	200 x 75 F7	450 Ø	2000
1600 mm	150 UC 23	1200 mm	200 x 75 F7	450 Ø	2400
1800 mm	150 UC 23	1200 mm	200 x 75 F7	450 Ø	2800

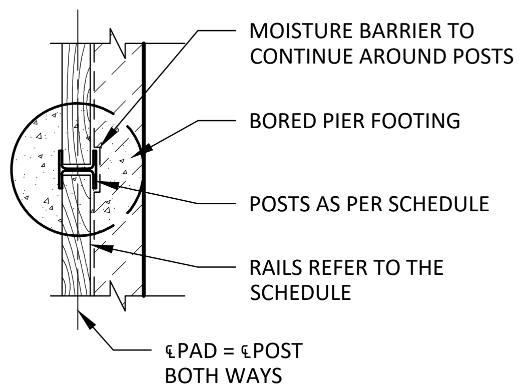
CONSTRUCTION SEQUENCE:-

1. BULK EXCAVATE SITE TO ALLOW SAFE BATTER SLOPE INSIDE PROPERTY BOUNDARIES IN ACCORDANCE WITH GEOTECHNICAL ENGINEERS RECOMMENDATIONS.
2. DRILL BORED PIER HOLES AS REQUIRED AND ENSURE HOLES ARE CLEAN OF LOOSE SOIL AND DEBRIS.
3. CAST IN VERTICAL GALVANISED POST TO THE APPROPRIATE DEPTH AND ALIGNMENT.
4. POUR FOOTING UP TO 100mm BELOW PROPOSED CUT LEVEL.
5. IF REQUIRED, SAFELY PROP EXISTING FENCE POST FOUNDATION UNTIL BACK OF WALL HAS BEEN BACKFILLED TO ENSURE THAT EXISTING FENCE FOUNDATION PADS DO NOT MOVE OUT OF VERTICAL ALIGNMENT
6. CAREFULLY EXCAVATE BETWEEN TWO POSTS AT ONE TIME AND INSERT AGGI PIPE AND/OR STRIP DRAIN, RETAINING SLEEPERS IMMEDIATELY AFTER EXCAVATION. THEN BACKFILL WITH COMPACTED FREE DRAINING MATERIAL, OR NO-FINES CONCRETE IF RETAINING WALL OFFSET IS LESS THEN 300mm FROM SITE BOUNDARY, AS SHOWN ON TYPICAL DETAIL.
7. REPEAT ITEMS 2-5 ONLY WHEN PREVIOUSLY RETAINING BAY HAS BEEN FULLY CONSTRUCTED AND BACKFILLED. AT NO STAGE ARE TWO ADJACENT BAYS BETWEEN POSTS TO BE CONSTRUCTED AT THE SAME TIME.
8. BUILDER CAN WORK ON DIFFERENT BAYS BETWEEN POSTS IF REQUIRED, PROVIDED AT LEAST TWO UN-DISTURBED BAYS EXIST BETWEEN EXCAVATED BAYS.

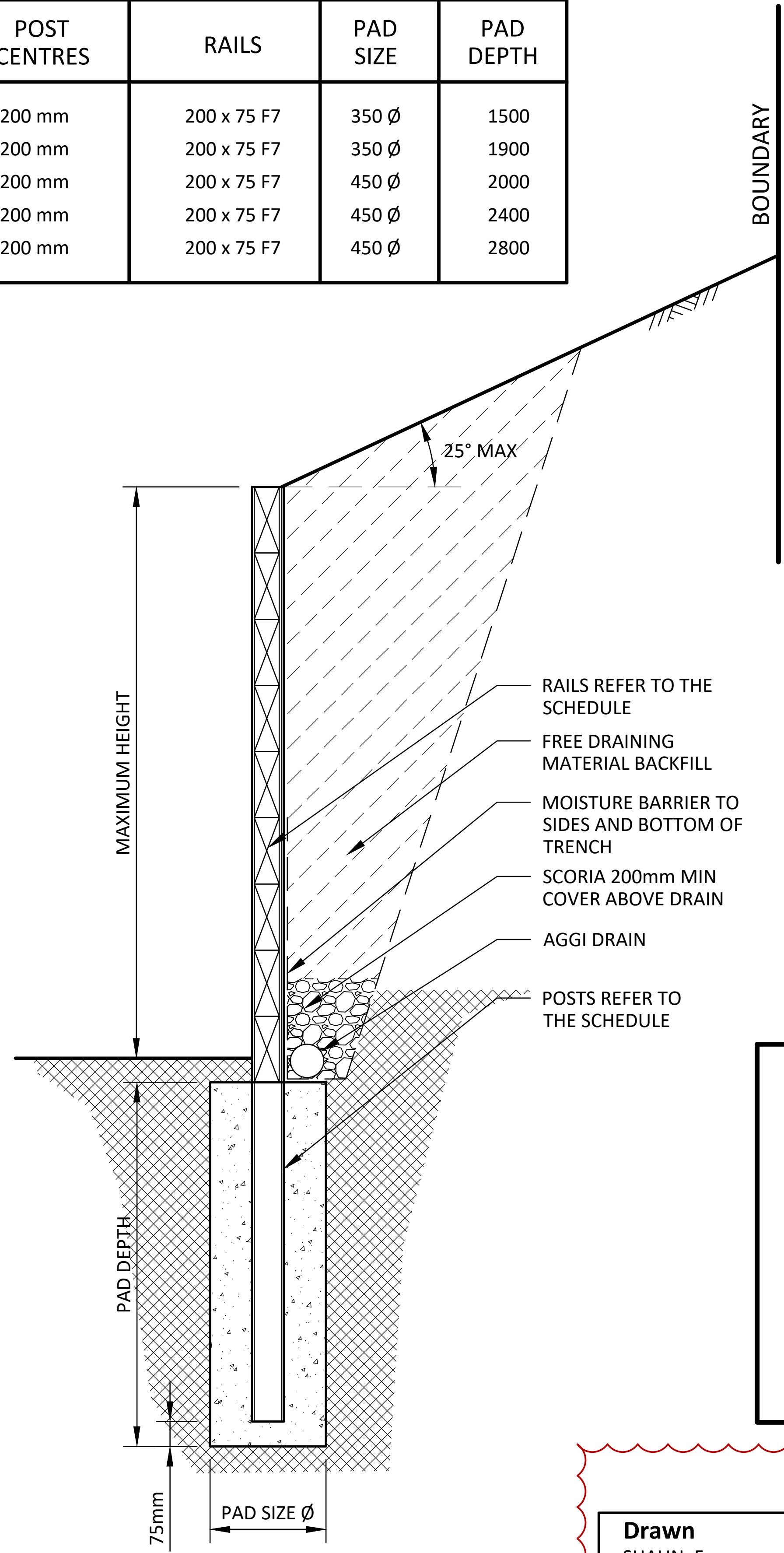


TYPICAL SLEEPER WALL PLAN

SCALE 1:50



PLAN:- STEEL POST



RETAINING WALL 2 NOTES:-

ALL POST PADS ARE TO BE FOUNDED FULL DEPTH AS NOTED IN THE SCHEDULE INTO THE **UNDERLYING NATURAL STIFF TO VERY STIFF CLAY. (200 KPa.)**

PADS AND A.G. DRAINS MUST NOT ENCROACH THE TITLE BOUNDARY.

RETAINING WALLS HAVE BEEN DESIGNED FOR A MAXIMUM SURCHARGE OF **5.0 KPa.**

FOR MAXIMUM RETAINING WALL DESIGN HEIGHT REFER TO SCHEDULE, SHOULD THE MAXIMUM HEIGHT BE EXCEEDED THE ENGINEER IS TO BE NOTIFIED FOR FURTHER ADVICE.

2

SLEEPER RETAINING WALL 2

Drawn
SHAUN .F
Design
MANA .M
Date
13/09/21
Scale---

Retaining Wall Details

Sheet No. **6**

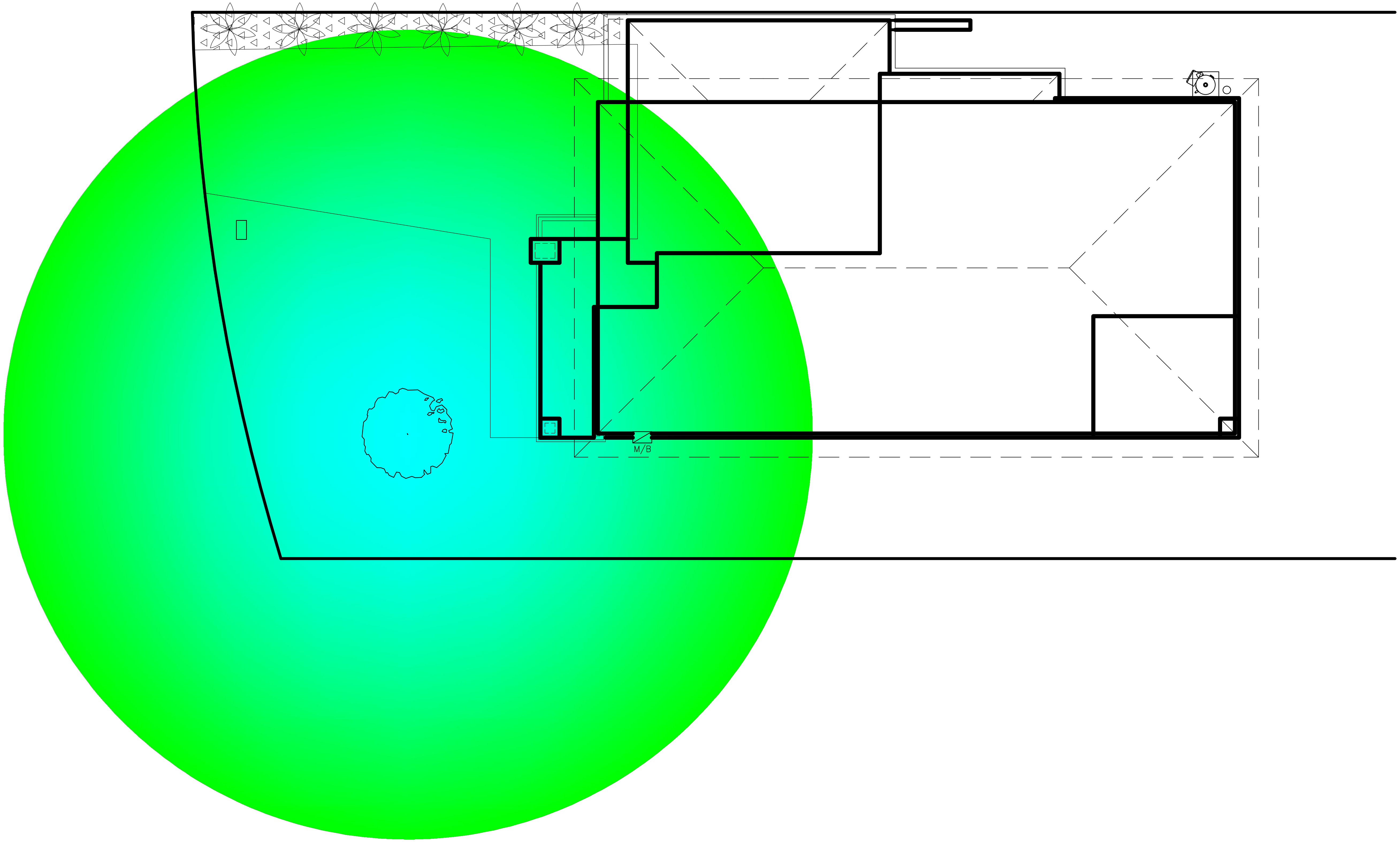
Job No: **2214421/2**

SIMONDS HOMES EAST MELBOURNE

LOT 31, OSKAR COURT, PAKENHAM

ACN 162 735 917
1st FLOOR, 25-27 RAILWAY RD,
BLACKBURN, VIC. 3130
E: contact@clickeng.com.au
P: (03) 9894 4222

CLICK ENGINEERING



"PLANTING OF TREES SHOULD BE AVOIDED NEAR A FOUNDATION OF A BUILDING OR NEIGHBOURING BUILDING ON REACTIVE CLAY SITES AS THEY CAN CAUSE DAMAGE DUE TO DRYING OF THE CLAY AT SUBSTANTIAL DISTANCES. TO REDUCE, BUT NOT ELIMINATE, THE POSSIBILITY OF DAMAGE, TREE PLANTING SHOULD BE RESTRICTED TO A DISTANCE FROM THE THE HOUSE AS FOLLOWS:

- *1.5 x MATURE HEIGHT FOR CLASS E SITES.*
- *1 x MATURE HEIGHT FOR CLASS H1 AND H2 SITES .*
- *0.75 x MATURE HEIGHT FOR CLASS M SITES."*

AUSTRALIAN STANDARD 2870-2011 APPENDIX B2.3 c

CURRENT ZONE OF INFLUENCE OF DRYING EFFECT ON CLAY SOILS DUE TO EXISTING TREES

EXPECTED ZONE OF INFLUENCE OF DRYING EFFECT ON CLAY SOILS DUE TO MATURE TREES

! IMMATURE TREE - GROWTH TO BE MONITORED. TREE REMOVAL OR PLACEMENT OF TREE ROOT BARRIERS MAY BE REQUIRED IN THE FUTURE, REFER TO ADJACENT NOTE.

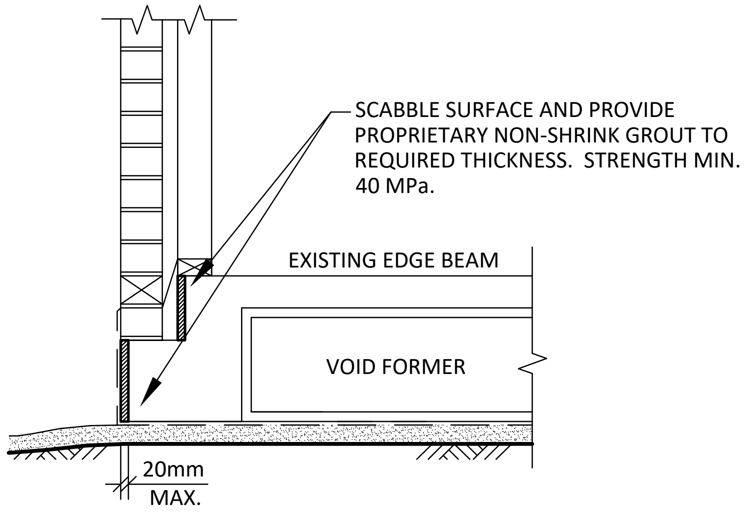
TREE INFLUENCE DIAGRAM

Drawn SHAUN .F	Sheet No. **7**
Design MANA .M	Job No: **2214421/2**
Date 13/09/21	SIMONDS HOMES EAST MELBOURNE
Scale 1:100	**LOT 31, OSKAR COURT, PAKENHAM**

Zone Of Influence Diagram

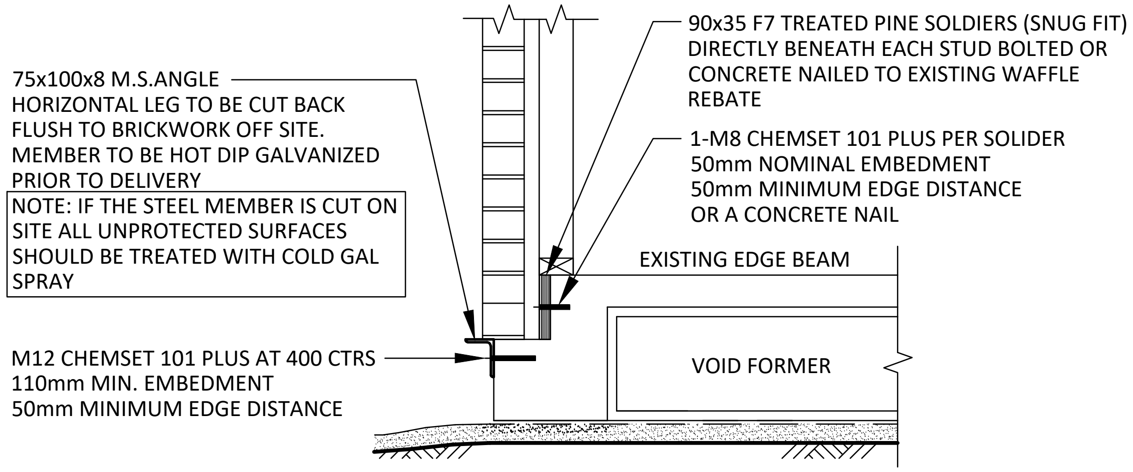
ACN 162 735 917
1st FLOOR, 25-27 RAILWAY RD,
BLACKBURN, VIC. 3130
E: contact@clickeng.com.au
P: (03) 9894 4222

CLICK ENGINEERING



WAFFLE RECTIFICATION
15-20mm MAX. OVERHANG

SCALE 1:20



WAFFLE RECTIFICATION
30mm MAX. OVERHANG

SCALE 1:20

NOTE:

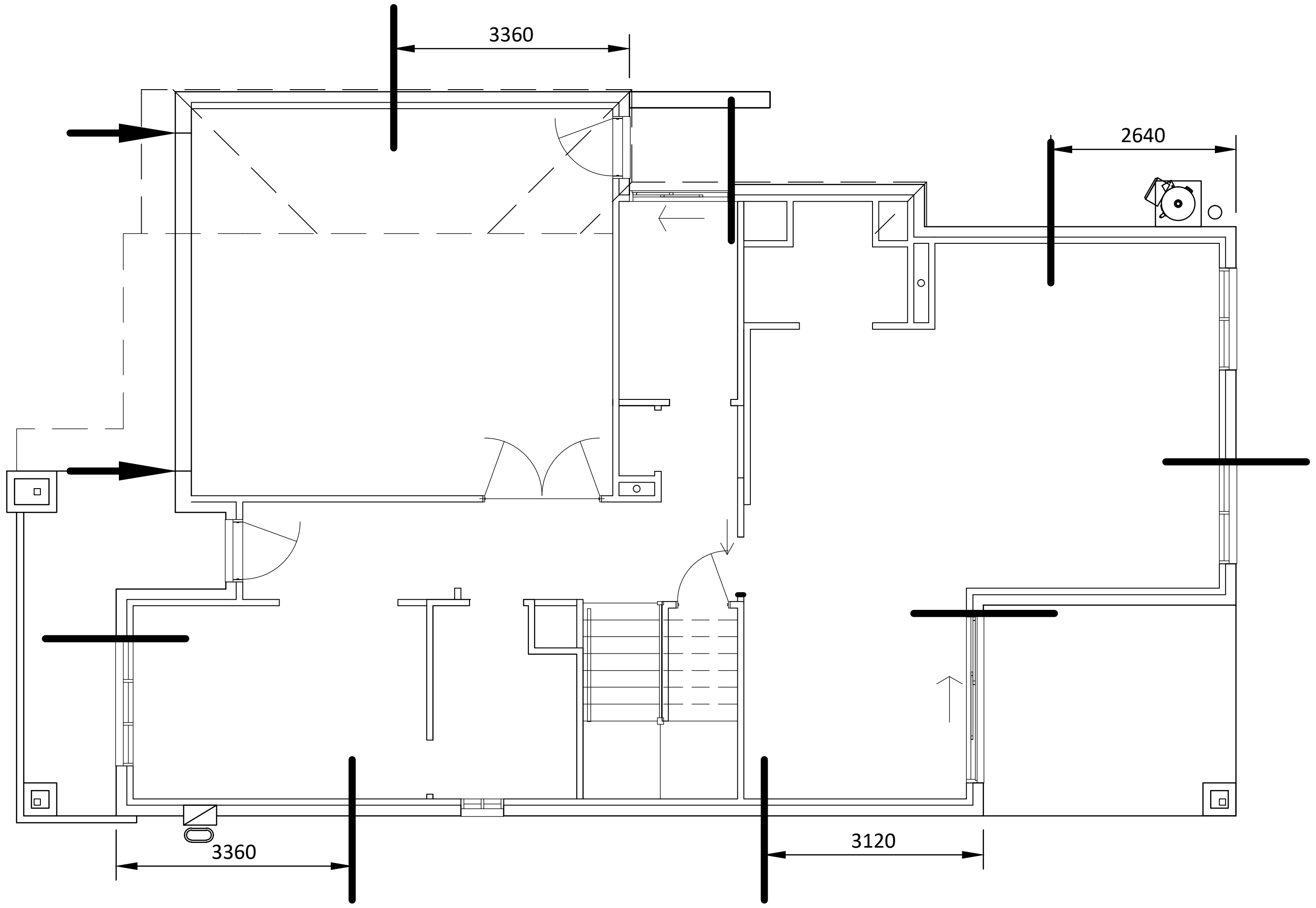
These standard overhang details may be adopted providing none of the below listed items have been breached. If one or more of the items have been breached this office should be contacted to provide a SITE SPECIFIC DESIGN SOLUTION.

The standard overhang details may be used providing the below items have been addressed.

- No 'Honeycombing' of the concrete is allowed to exist on the side of the edge beam (ie voids due to a lack of vibrating during the concrete pour or stripping of formwork)
- The overhang must not exist in the vicinity of a concentrated load (ie Multiple Stud or Steel Column Support)
- A minimum of 90mm timber framing must be used. The connection to the slab should not be restricted by the overhang and should still be in accordance with the Timber Framing Manual (AS1684) for the relevant wind terrain category.
- Details can be adopted under braced walls where nominal fixing is required. Specific tie downs in N3 Wind terrain areas will need to be looked at individually.
- Brick cavity sizes can be manipulated to reduce or avoid overhang however the minimum cavities should be in accordance with the current National Construction Code

Drawn SHAUN .F **Design** MANA .M **Date** 13/09/21 Scale 1:20	Overhang Details	Sheet No. **8** Job No: **2214421/2** SIMONDS HOMES EAST MELBOURNE **LOT 31, OSKAR COURT, PAKENHAM**

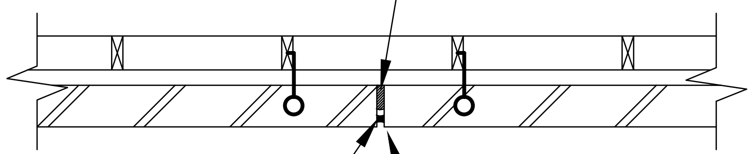
ACN 162 735 917
1st FLOOR, 25-27 RAILWAY RD,
BLACKBURN, VIC. 3130
E: contact@clickeng.com.au
P: (03) 9894 4222



PROVIDE SUPPORT FOR WALL PANELS BY FIXING CAVITY WALL TIES TO EVERY FOURTH COURSE ON EACH SIDE OF ARTICULATION JOINTS.

NOTE:
ARTICULATION JOINTS TO BE IN ACCORDANCE WITH THE CEMENT CONCRETE & AGGREGATES AUSTRALIA TECHNICAL NOTE 61.

COMPRESSIBLE FOAM JOINT FILLER AND MASTIC BACKING

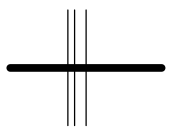


MASTIC SEALANT TO BE PACKED OR GUNNED INTO THE JOINT

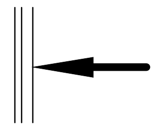
10-20mm GAP BETWEEN BRICKWORK

MASONRY ARTICULATION JOINT DETAIL

LEGEND:



BRICKWORK ARTICULATION JOINT LOCATION



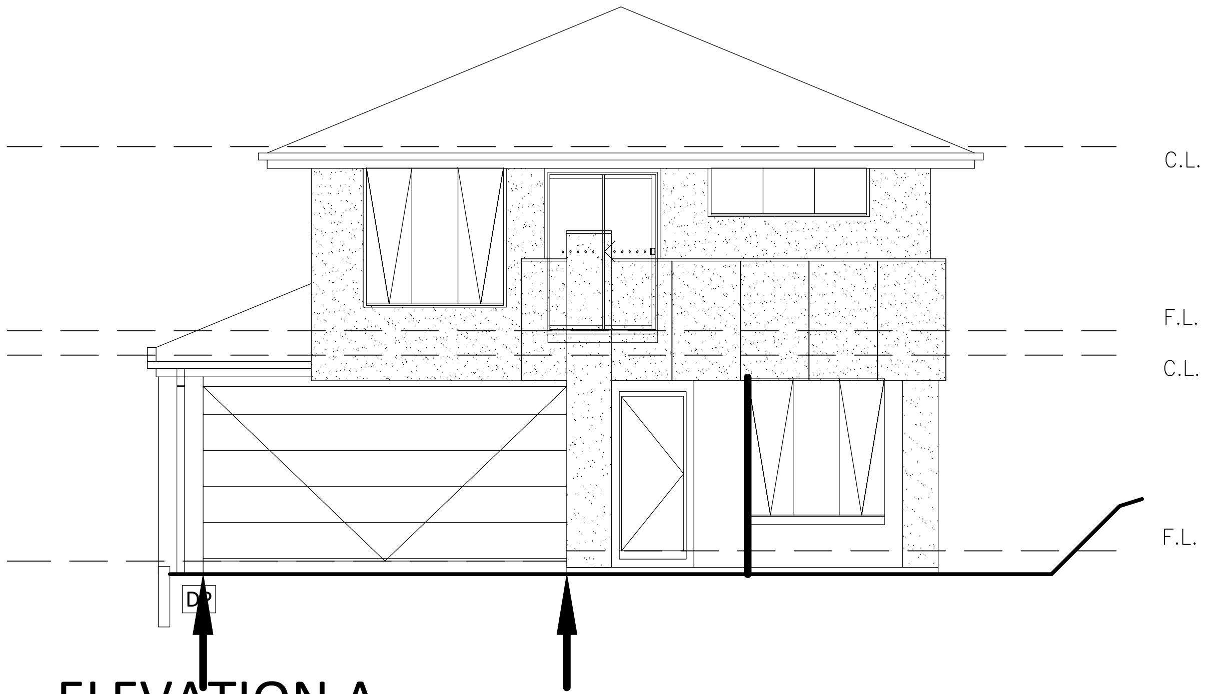
NATURAL ARTICULATION JOINT LOCATION i.e. FULL HEIGHT DOOR AND/OR WINDOW OPENING OR CLADDING/BRICKWORK INTERFACE

BRICKWORK ARTICULATION PLAN

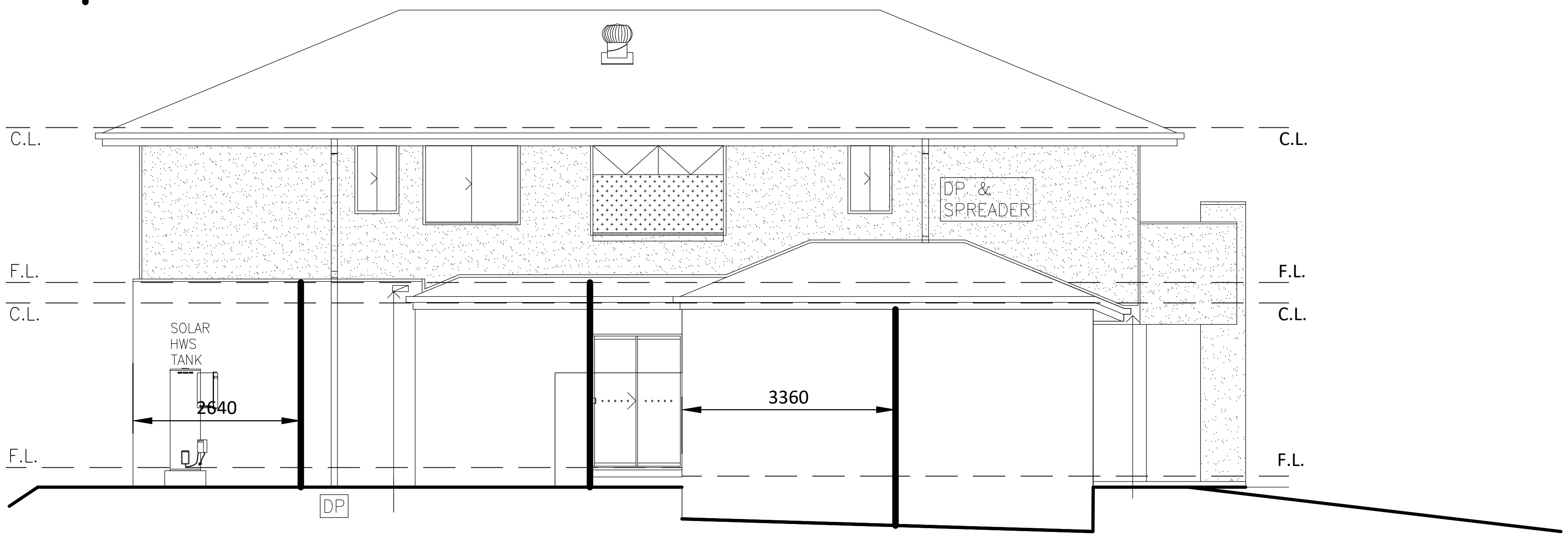
Drawn SHAUN .F	Sheet No. **9**
Design MANA .M	Job No: **2214421/2**
Date 13/09/21	SIMONDS HOMES EAST MELBOURNE
Scale 1:100	**LOT 31, OSKAR COURT, PAKENHAM**
Articulation Plan	

ACN 162 735 917
1st FLOOR, 25-27 RAILWAY RD,
BLACKBURN, VIC. 3130
E: contact@clickeng.com.au
P: (03) 9894 4222

CLICK ENGINEERING



ELEVATION A



ELEVATION B

Drawn
SHAUN .F
Design
MANA .M
Date
13/09/21
Scale 1:100

Articulation Elevations

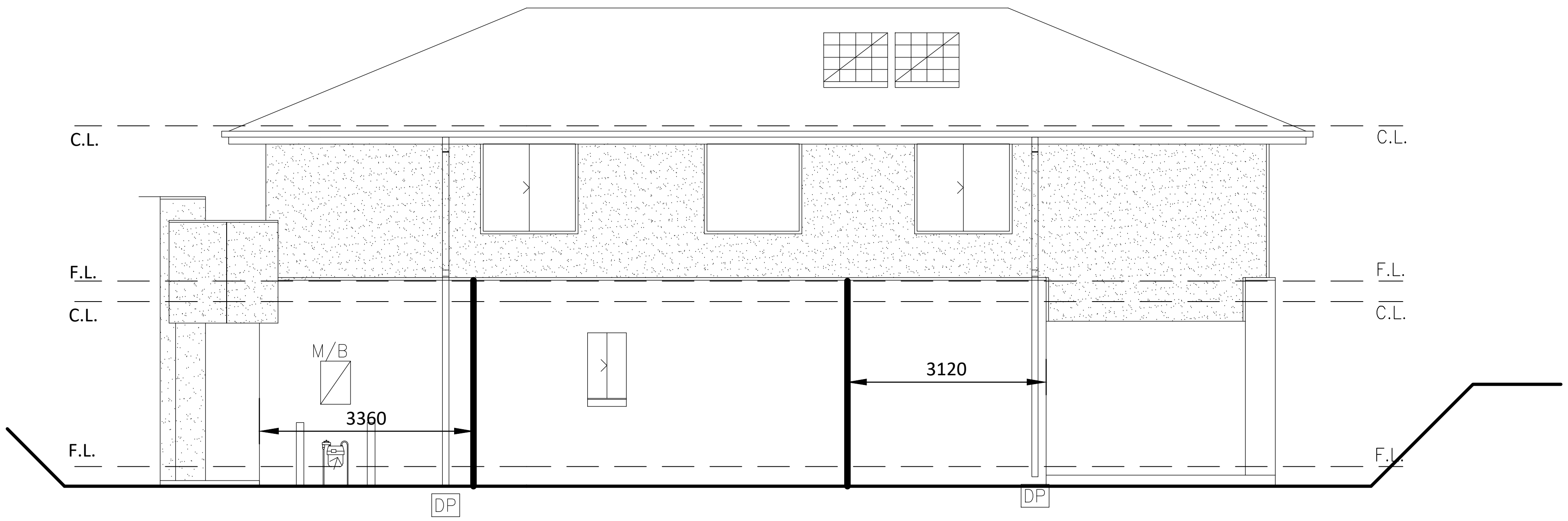
Sheet No. **10**

Job No: **2214421/2**
SIMONDS HOMES EAST MELBOURNE
LOT 31, OSKAR COURT, PAKENHAM

ACN 162 735 917
1st FLOOR, 25-27 RAILWAY RD,
BLACKBURN, VIC. 3130
E: contact@clickeng.com.au
P: (03) 9894 4222

ENGINEERING

ELEVATION C



ELEVATION D

Drawn
SHAUN .F
Design
MANA .M
Date
13/09/21
Scale 1:100

Articulation Elevations

Sheet No. **11**

Job No: **2214421/2**
SIMONDS HOMES EAST MELBOURNE
LOT 31, OSKAR COURT, PAKENHAM

ACN 162 735 917
1st FLOOR, 25-27 RAILWAY RD,
BLACKBURN, VIC. 3130
E: contact@clickeng.com.au
P: (03) 9894 4222

CLICK ENGINEERING

TIMBER NOTES:-

GENERAL:-
ALL TIMBER FRAMING, TYPE, FIXINGS AND BRACING ETC. TO BE IN ACCORDANCE WITH AS. 1684-2010 (LIGHT TIMBER FRAMING MANUAL) AND ANY SUPPLEMENTS AND/OR THE ARCHITECTS SPEC's.

DOUBLE STUDS:-
PROVIDE 2-90 x 45 MGP10 STUDS LOCATED UNDER ENDS OF ALL LINTELS, ROOF BEAMS, GIRDER TRUSSES, FLOOR BEAMS AND AT SIDES OF WINDOW OPENINGS OF ALL MEMBERS SPECIFIED ON THIS DOCUMENTATION UNLESS NOTED OTHERWISE.

POINT LOADS:-
PROVIDE MULTIPLE STUD SUPPORTS OF SAME SIZE, NUMBER AND TYPE AS THE SUPPORT FROM ABOVE UNLESS NOTED OTHERWISE ON PLAN.

MULTIPLE MEMBERS:-
MULTIPLE TIMBER MEMBERS ARE TO BE SPIKED TOGETHER TO FORM A SINGLE UNIT. (REFER TO THE TIMBER FRAMING MANUAL.)

WALL BRACING:-
ALL WALL BRACING, TYPE, POSITION AND FIXING TO BE IN ACCORDANCE WITH AS. 1684-2010.

TIE DOWNS:-
TYING DOWN OF ALL TIMBER MEMBERS INCLUDING ROOF FRAMING TO BE IN ACCORDANCE WITH AS. 1684-2010.

COLUMNS:-
ALL INTERNAL STEEL COLUMNS ARE TO BE FIXED TO THE ADJACENT STUD WALLS WITH M10 BOLTS OR SUITABLE IMPACT FASTENERS AT TOP AND BOTTOM OF THE COLUMNS AND AT NOGGING CENTRES.

METAL FASTENERS:-
ALL NOMINATED METAL FASTENERS (OR THEIR EQUIVALENT) ARE TO BE FIXED IN ACCORDANCE WITH THE MANUFACTURERS DETAILS AND/OR SPECIFICATIONS.

INTERNAL N.L.B. WALLS:-
ALL INTERNAL NON LOAD BEARING WALLS ARE TO BE FIXED TO THE RAFTERS OR FLOOR JOISTS WITH SUITABLE APPROVED FLEXIBLE CONNECTORS.

ROOF BATTENS:-
TIMBER ROOF BATTENS TO ARCHITECTS DETAILS AND/OR SPECIFICATIONS.

DURABILITY:-
ALL EXTERNAL ABOVE GROUND EXPOSED TIMBERS ARE TO BE PRESERVATIVE TREATED TO H3 LEVEL OR EQUIVALENT NATURAL DURABLE HARDWOOD ALTERNATIVES USE.

BRICKWORK LINTELS

HEIGHT OVER LINTEL(mm)	CLEAR SPAN OF OPENING(mm)							
BRICK WORK	1000	1200	1500	1800	2100	2400	2700	3000
500	75x100x6	75x100x6	75x100x6	75x100x6	75x100x6	100x100x6	100x100x6	100x100x6
1000	75x100x6	75x100x6	75x100x6	75x100x6	100x100x6	100x100x6	100x100x6	150x100x10
1500	75x100x6	75x100x6	75x100x6	100x100x6	100x100x6	100x100x6	150x100x10	150x100x10
2000	75x100x6	75x100x6	100x100x6	100x100x6	100x100x6	150x100x10	150x100x10	150x100x10
2500	75x100x6	75x100x6	100x100x6	100x100x6	100x100x6	150x100x10	150x100x10	150x100x10
3000	75x100x6	75x100x6	100x100x6	100x100x6	100x100x6	150x100x10	150x100x10	

NOTE:- FIRST DIMENSION CORRESPONDS TO THE VERTICAL LINTEL LEG. e.g. 75x100x6LINTEL. 75mm LEG VERTICAL

STEEL MEMBERS:-

ALL EXTERNAL STEEL MEMBERS ARE TO BE HOT DIP GALVANIZED OR FINISHED IN ACCORDANCE WITH THE ARCHITECTS DETAILS AND/OR SPECIFICATIONS.

ALL STEEL MEMBERS TO HAVE 150mm MIN. END BEARING ONTO BRICKWORK AT EACH END UNLESS NOTED OTHERWISE IN THE SCHEDULE OR OTHERWISE DETAILED.

INVERTED "T" LINTELS TO BE WELDED TOGETHER. 6FW. 200 MIN. EACH END THEN WELD 150 MISS 150 STAGGERED BOTH SIDES FULL LENGTH.

COMPOSITE MEMBERS (CHANNEL AND ANGLE) TO BE WELDED TOGETHER. 6FW. 200 MIN. EACH END THEN WELD 150 MISS 150 STAGGERED BOTH SIDES FULL LENGTH.

CRANKED OR CORNER LINTELS ARE TO BE MITRE CUT AND FULL STRENGTH BUTT WELDED TOGETHER AT CRANKS OR CORNERS.

ANY SITE WELDING OF GALVANIZED MEMBERS TO BE PREPARED AND COLD-GALV. FINISH APPLIED IN STRICT ACCORDANCE WITH THE MANUFACTURERS SPECIFICATIONS.

TEMPORARY WALL BRACING:-

THE BUILDER/CARPENTER IS TO PROVIDE ADEQUATE TEMPORARY BRACING TO PLUMB WALLS AT EACH LEVEL DURING CONSTRUCTION. TEMPORARY BRACING IS ADDITIONAL TO ANY BRACING SHOWN ON THESE DRAWINGS AND MUST REMAIN IN POSITION UNTIL STUD FRAMING, ALL PERMANENT WALL BRACING, STEELWORK AND THE PLATFORM FLOOR AND ROOF OVER HAS BEEN COMPLETED.

NOTE:-

THE ENGINEERING DETAILS PROVIDED ON THIS PROJECT ARE INDICATIVE OF WHAT IS REQUIRED AND ARE NOT SHOP DRAWINGS. FABRICATORS MUST ENSURE DETAILS PROVIDED ARE CHECKED AGAINST SITE CONDITIONS PRIOR TO ANY FABRICATION/ERECTION. DETAILS MAY NEED TO BE AMENDED TO SUIT SITE CONDITIONS. THE DESIGN ENGINEER SHOULD BE CONTACTED TO CONFIRM AMENDMENTS AND/OR REDESIGN.

ACN 162 735 917
1st FLOOR, 25-27 RAILWAY RD,
BLACKBURN, VIC. 3130
E: contact@clickeng.com.au
P: (03) 9894 4222

CLICK ENGINEERING

DOUBLE STUDS:-
PROVIDE 2-90 x 45 MGP10 STUDS LOCATED UNDER ENDS OF ALL MEMBERS SPECIFIED ON THIS DOCUMENTATION UNLESS NOTED OTHERWISE.
ALL BEAMS AND SUPPORTS NOT SPECIFIED ON THIS DOCUMENTATION ARE TO BE DESIGNED BY OTHERS.

POINT LOADS:-
POINT LOADS FOUNDED DIRECTLY ONTO WAFFLE SLAB PANELS MUST NOT EXCEED 10kN SAFE WORKING LOAD.
IF SO, THIS OFFICE SHOULD BE CONTACTED FOR FURTHER CONSULTING.

ROOF TRUSSES:-
PREFAB TIMBER ROOF TRUSSES, DESIGN, CENTRES, FIXING AND BRACING BY OTHERS. GIRDER TRUSS LOCATIONS HAVE BEEN ASSUMED AND MUST BE CONFIRMED WITH FINAL TRUSS LAYOUT DRAWINGS TO CONFIRM THE SUPPORTING MEMBER SIZES.

NOTE:
ALL BEAMS TO HAVE 90mm MIN. END BEARING UNLESS SPECIFIED OTHERWISE.

NOTE:-
THE DESIGN/DOCUMENTATION BY CLICK ENGINEERING HAS BEEN LIMITED TO THE REQUESTS OF THE CLIENT. ANY ITEMS NOT SHOWN ON THESE DRAWINGS TO BE BY THE BUILDING DESIGNER OR OTHER CONSULTANTS.

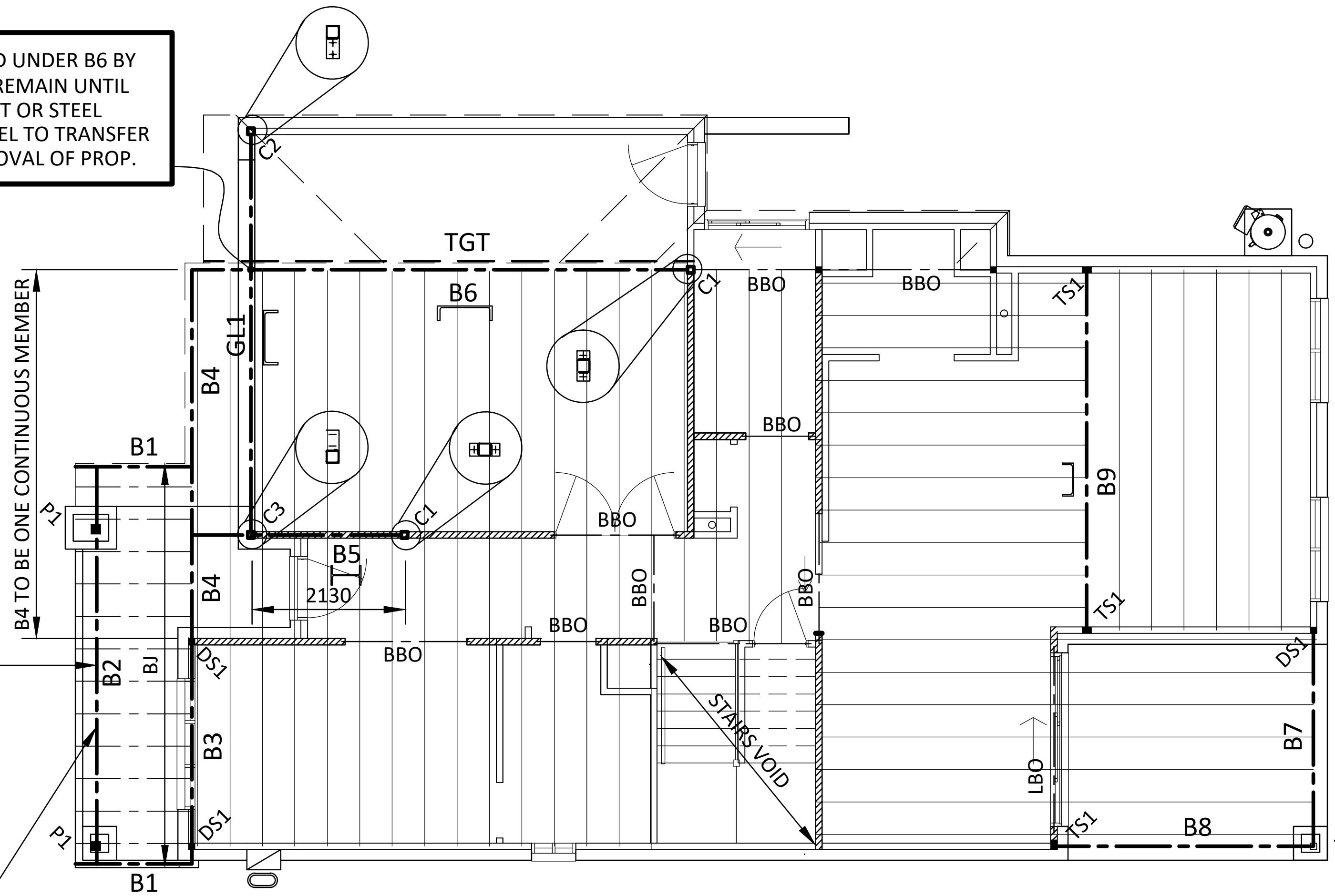
FLOOR JOISTS:-
FLOOR JOISTS, HANDLING, SETOUT, PLACEMENT, FIXING, CONNECTIONS, STRONGBACKS, BRACING, BLOCKING AND TRIMMERS ETC. TO BE IN ACCORDANCE WITH THE MANUFACTURERS SPECIFICATIONS. MANUFACTURER TO CONFIRM JOIST TYPE AND SIZES.

LBW - LOAD BEARING WALL

MEMBER SCHEDULE	
TRUSSES	TIMBER ROOF TRUSSES @ 600 CTRS TO MANUFACTURERS SPECIFICATIONS
TGT	TRUNCATED GIRDER TRUSS (ASSUMED LOCATIONS)
B1	190 x 45 F7 KD TREATED PINE
B2	2/300 x 63 H3 TREATED HYSPAN LVL (BULKHEAD) OR 3/290 x 45 H3 TREATED HYSPAN LVL
B3	2/290 x 45 HYSPAN LVL
B4	2/290 x 45 HYSPAN LVL
B5	250 UB 26
B6	380 PFC
B7	2/290 x 45 HYSPAN LVL
B8	2/290 x 45 HYSPAN LVL
B9	230 PFC
BB0	BEAM BY OTHERS
GL1	380 PFC
BJ	190 x 45 F7 KD TREATED PINE BALCONY JOISTS @ 450 CTRS.
C1	89x89x5.0 SQUARE HOLLOW SECTION POST
C2	89x89x5.0 SQUARE HOLLOW SECTION POST
C3	89x89x6.0 SQUARE HOLLOW SECTION POST
P1	135 x 135 F7 KD PINE POST
P2	135 x 135 F7 KD PINE POST
DS1	2/90x45 F17 KD HARDWOOD
TS1	3/90x45 F17 KD HARDWOOD

NOTE: GARAGE LINTEL, TO BE PROPPED UNDER B6 BY 4/90x45 F17 KDHW STUDS AND IS TO REMAIN UNTIL BRICKWORK IS UP, STRUCTURAL GROUT OR STEEL PACKERS ARE TO BE USED UNDER LINTEL TO TRANSFER LOADS TO BRICKWORK PRIOR TO REMOVAL OF PROP.

NOTE: B2 TO BE PROPPED MIDSPAN BY 2/90x45 F17 KDHW STUDS AND IS TO REMAIN UNTIL BRICKWORK IS UP, STRUCTURAL GROUT OR STEEL PACKERS ARE TO BE USED UNDER LINTEL TO TRANSFER LOADS TO EACH SIDE OF BRICKPIERS PRIOR TO REMOVAL OF PROP.



PROVIDE TREATED SOLID BLOCKING ABOVE BEAM B2 BETWEEN THE JOISTS JOISTS TO BE TIED DOWN WITH GALVANIZED MULTIGRIPS TO B2

LOWER ROOF & FIRST FLOOR SUPPORT PLAN

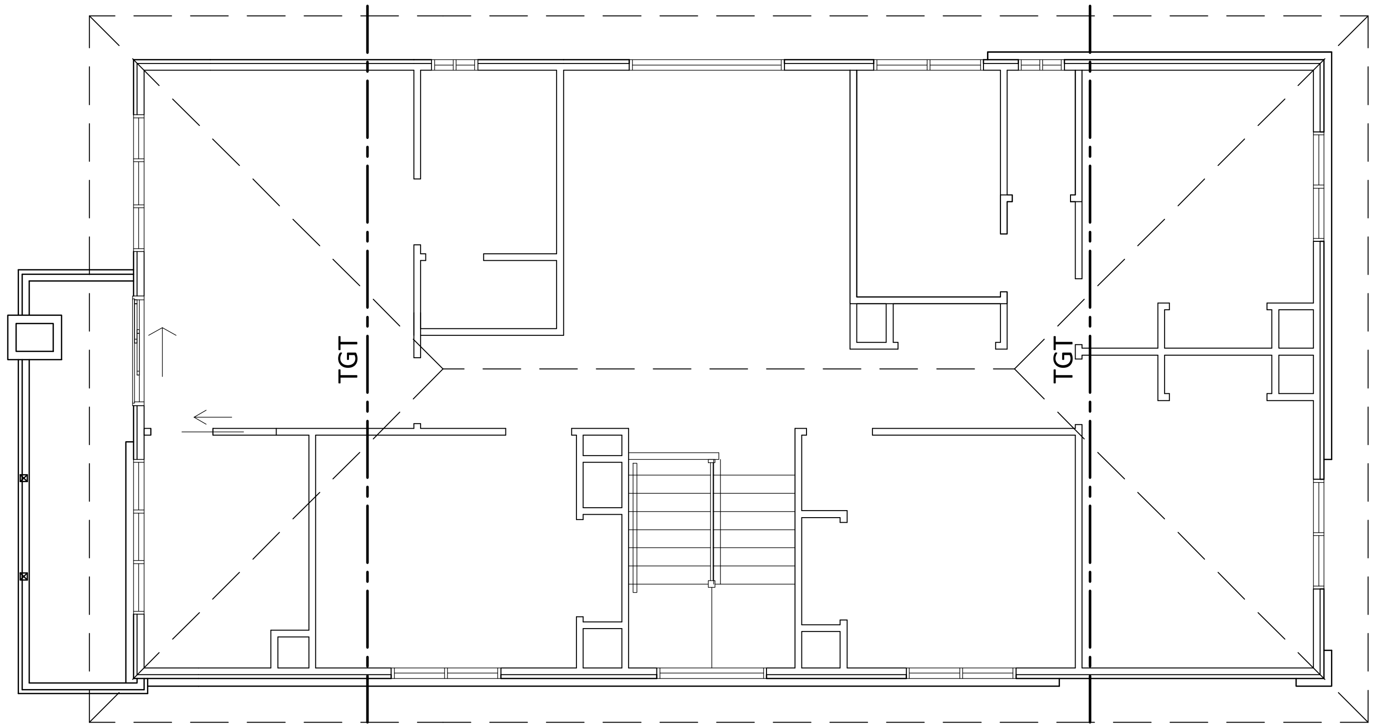
Drawn SHAUN .F
Design MANA .M
Date 13/09/21
Scale 1:100

Lower Roof & First Floor Support Plan

Sheet No. **13**

Job No: **2214421/2**
SIMONDS HOMES EAST MELBOURNE
LOT 31, OSKAR COURT, PAKENHAM

ACN 162 735 917
1st FLOOR, 25-27 RAILWAY RD, BLACKBURN, VIC. 3130
E: contact@clickeng.com.au
P: (03) 9894 4222



ROOF SUPPORT PLAN

Drawn
SHAUN .F
Design
MANA .M
Date
13/09/21
Scale 1:100

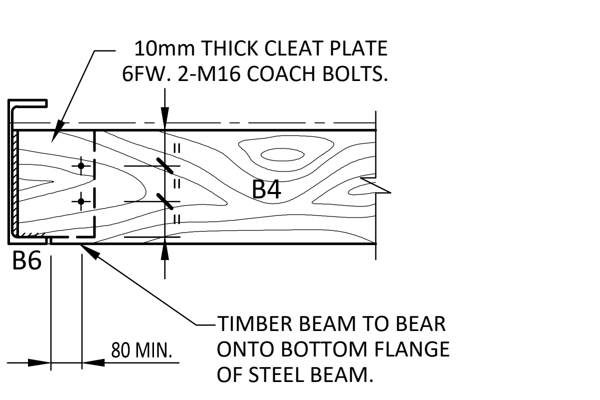
Roof Support Plan

Sheet No. **14**

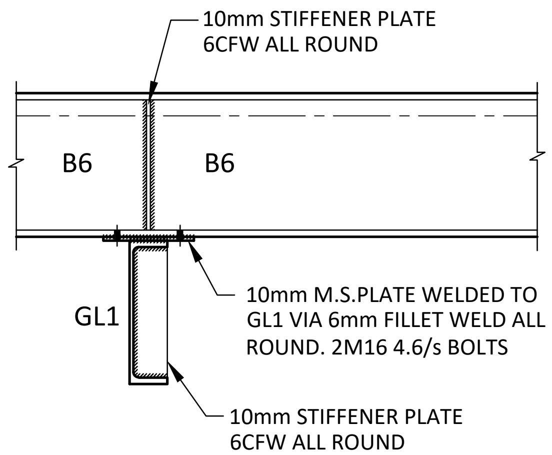
Job No: **2214421/2**
SIMONDS HOMES EAST MELBOURNE
LOT 31, OSKAR COURT, PAKENHAM

ACN 162 735 917
1st FLOOR, 25-27 RAILWAY RD,
BLACKBURN, VIC. 3130
E: contact@clickeng.com.au
P: (03) 9894 4222

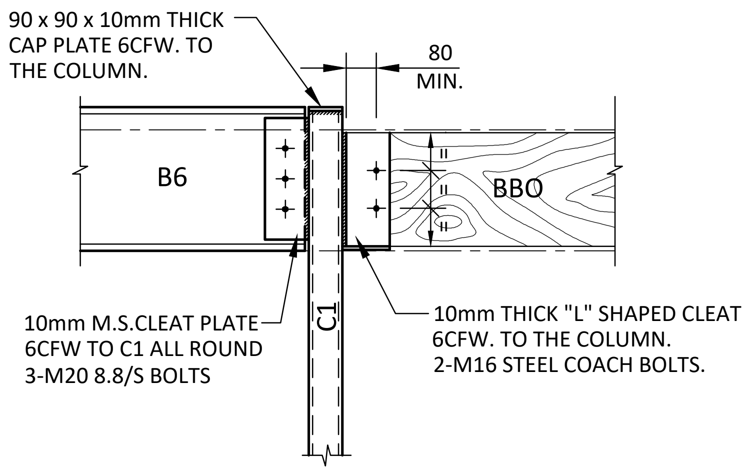
CLICK ENGINEERING



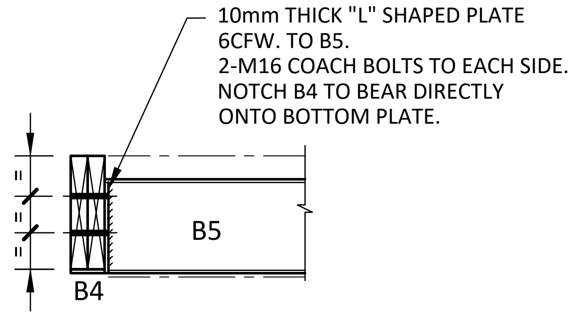
CONN:- B4/B6



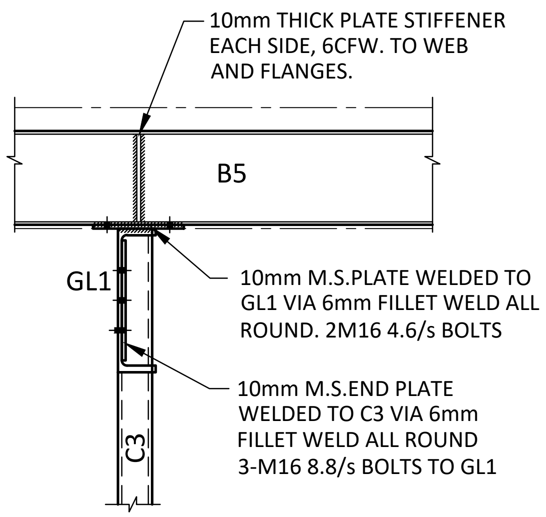
DETAIL- B6/GL1



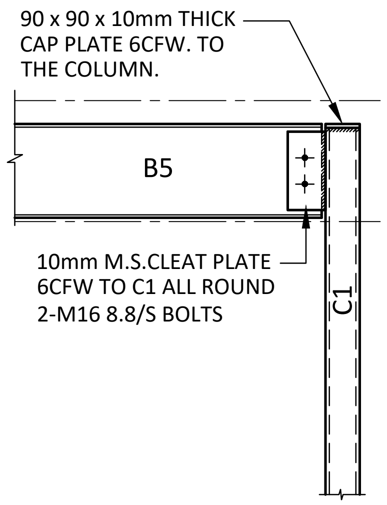
DETAIL- B6 & BBO/C1



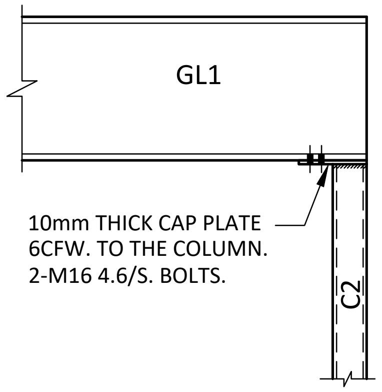
DETAIL- B4/B5



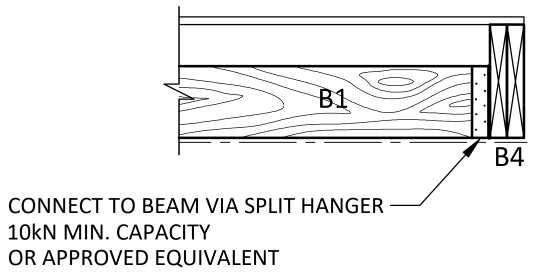
DETAIL- B5 & GL1/C3



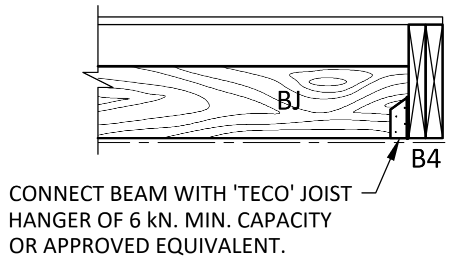
DETAIL- B5/C1



DETAIL- GL1/C2



CONN:- B1/B4



CONN:- BJ/B4

SIMILAR DETAIL: BJ/B3

Drawn
SHAUN .F
Design
MANA .M
Date
13/09/21
Scale 1:20

Structural Details

Sheet No. **15**

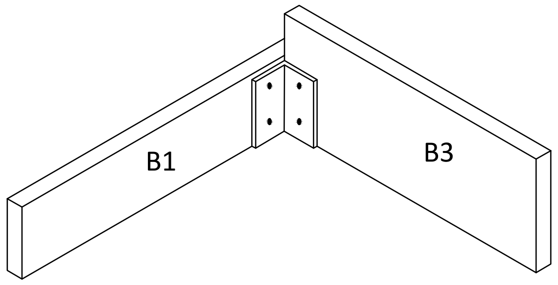
Job No: **2214421/2**

SIMONDS HOMES EAST MELBOURNE

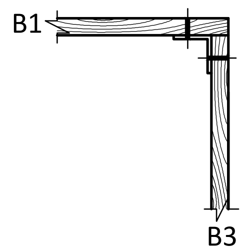
LOT 31, OSKAR COURT, PAKENHAM

ACN 162 735 917
1st FLOOR, 25-27 RAILWAY RD,
BLACKBURN, VIC. 3130
E: contact@clickeng.com.au
P: (03) 9894 4222

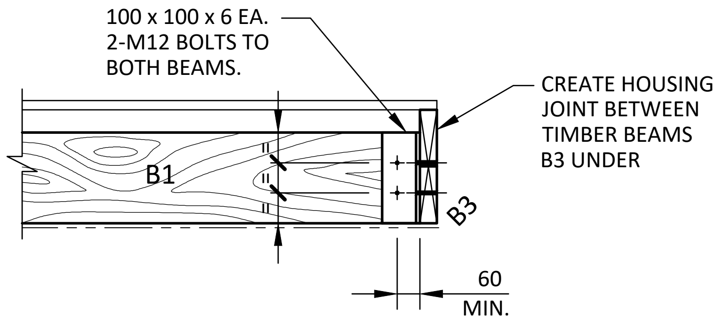
CLICK ENGINEERING



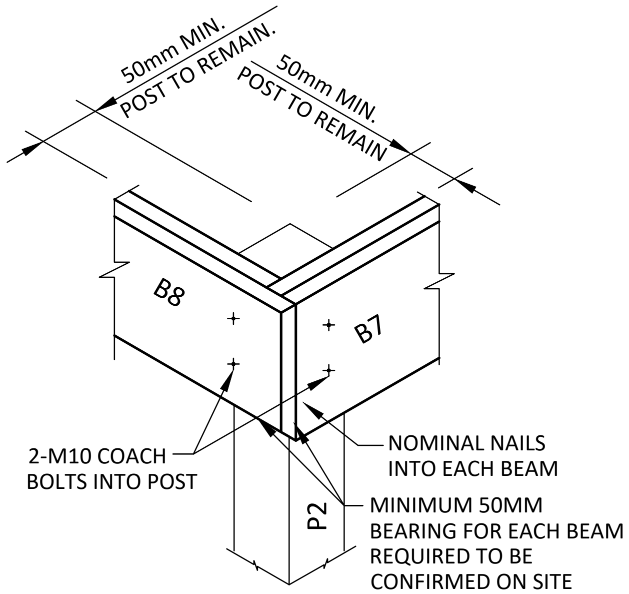
ISOMETRIC VIEW



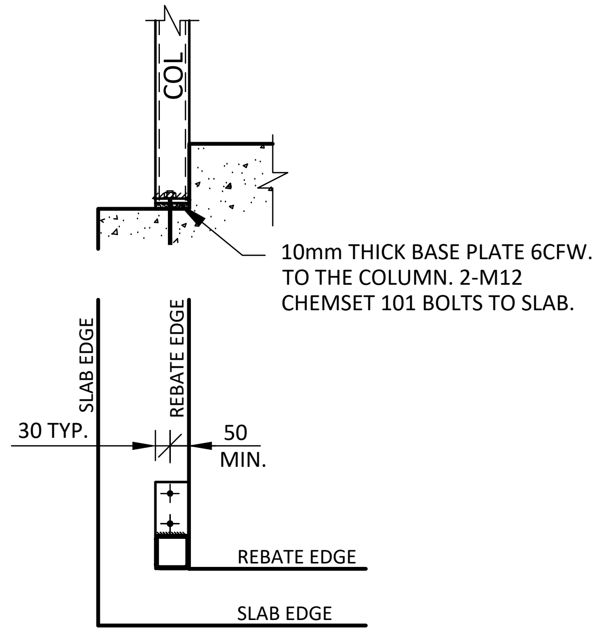
PLAN VIEW



CONN:- B1/B3



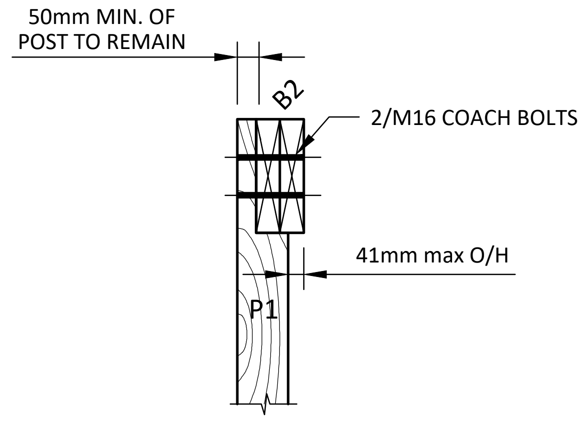
CONN:- B7&B8/P2



PLAN VIEW

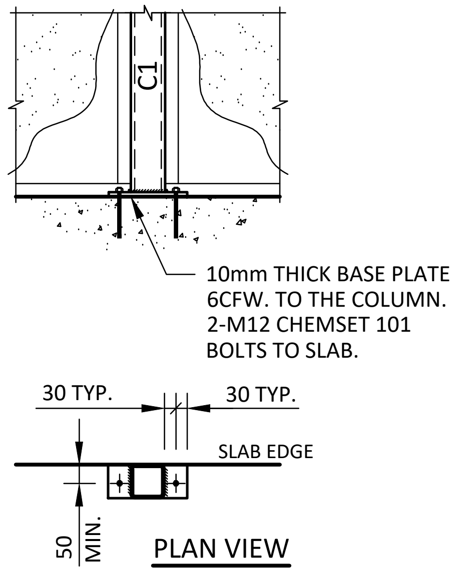
GARAGE DETAIL:- C2

SIMILAR DETAIL: C3



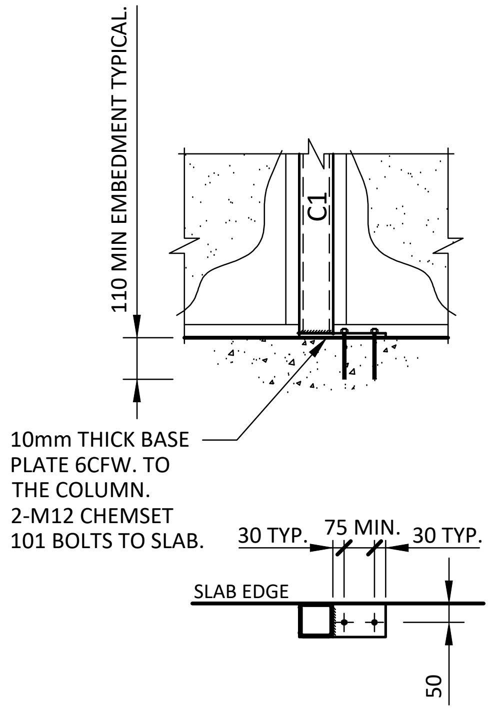
DETAIL- B2/P1

SCALE 1:20



PLAN VIEW

PERIMETER COLUMN:- C1



PLAN VIEW

PERIMETER C1 COLUMN

Drawn
SHAUN .F
Design
MANA .M
Date
13/09/21
Scale 1:20

Structural Details

Sheet No. **16**

Job No: **2214421/2**
SIMONDS HOMES EAST MELBOURNE
LOT 31, OSKAR COURT, PAKENHAM

ACN 162 735 917
1st FLOOR, 25-27 RAILWAY RD,
BLACKBURN, VIC. 3130
E: contact@clickeng.com.au
P: (03) 9894 4222

CLICK ENGINEERING

EROSION CONTROL:

1. THE BUILDER SHALL IMPLEMENT AN EFFECTIVE EROSION CONTROL PLAN TO ENSURE THAT THERE IS NO DISCHARGE OF SEDIMENT FROM THE SITE.
2. EXCAVATIONS AND EXPOSURE OF SOILS SHOULD ONLY OCCUR ON THE ACTUAL HOUSE PLATFORM, RATHER THAN THE ENTIRE SITE. VEGETATION INCLUDING GRASS SHOULD BE KEPT WHERE POSSIBLE, DURING THE CONSTRUCTION PERIOD.
3. ALL STORMWATER SHOULD BE RE-DIRECTED FROM THE CONSTRUCTION AREA VIA APPROPRIATE GRADING OF THE SITE AND/OR PLACEMENT OF OPEN DRAINS.
4. A FORMAL CONSTRUCTION ACCESS POINT SHOULD BE ESTABLISHED. FORMAL CROSSOVER PROTECTION SHOULD BE PLACED. A CRUSHED ROCK BASE, WITH APPROPRIATE DEPTH TO STOP SOILS FROM SURFACING, SHOULD BE PLACED FOR DELIVERY AND CONSTRUCTION VEHICLE ACCESS.
5. WHERE FILL IS TO BE STOCKPILED, THEN A SEDIMENT FENCE SHOULD BE PLACED AROUND THE FILL.
6. A SILT FENCE, HAY BALES OR SEDIMENT SANDS SHOULD BE PLACED ACROSS THE LOWEST POINT OF THE SITE IN ORDER TO STOP ANY SEDIMENT FROM LEAVING THE SITE, AND TO PROTECT ADJACENT SITES.
7. CONSTRUCTION VEHICLES, WHICH COLLECT SEDIMENT DUE TO THEIR PROCESS, EG BACKHOES, SHOULD BE WASHED DOWN BEFORE LEAVING THE SITE.
8. ANY SEDIMENT INADVERTENTLY LEAVING THE SITE, SHOULD BE COLLECTED FROM THE ROAD USING WASHING AND COLLECTION METHODS, PRIOR TO IT ENTERING A STORMWATER DRAIN.
9. THE STORMWATER INLET POINT SHOULD BE PROTECTED AGAINST SILT INGRESS VIA; SILT NETTING, HAY BALES OR ANY APPROVED PROPRIETY SYSTEM.

MIN. REQUIREMENTS FOR SEWER ARTICULATION

SITE CLASS	SEWER EXIT POINTS & ORG		RISERS	
	SWIVEL	EXPANDER	EXPANDER	LAGGING
'M'	0	0	NOTE 1	MIN. 20mm
'H1'	1	1	NOTE 1	MIN. 40mm
'H2'	2	1	NOTE 1	MIN. 40mm
'E'	2	1	NOTE 2	MIN. 40mm

NOTE 1 - MAX DEPTH OF SEWER TO UNDERSIDE OF SLAB TO BE 900mm FOR CLASS M & 600mm FOR CLASS H1 & H2, OTHERWISE EXPANDER JOINT REQUIRED.
NOTE 2 - EXPANDER REQUIRED ON ALL RISERS.

MIN. REQUIREMENTS EXPANSION JOINT CAPACITY

SITE CLASS	EXPANDER MOVEMENT RANGE	SWIVEL ROTATION RANGE
'E'	150mm	15°
'H1&H2'	70mm	15°
'M'	MIN. 20mm LAGGING THROUGH FOOTING	NOT APPLICABLE

DRAINAGE REQUIREMENTS:

1. SURFACE DRAINAGE OF THE SITE SHALL BE CONTROLLED FROM THE START OF SITE PREPARATION AND CONSTRUCTION. THE DRAINAGE SYSTEM SHALL BE COMPLETED BY THE FINISH OF CONSTRUCTION OF THE BUILDING.
2. SURFACE DRAINAGE SHALL BE CONSTRUCTED TO AVOID WATER PONDING AGAINST OR NEAR THE FOOTING. THE GROUND IN THE IMMEDIATE VICINITY OF THE PERIMETER FOOTING, INCLUDING THE GROUND UPHILL FROM THE SLAB ON CUT AND FILL SITES, SHALL BE GRADED TO FALL 50mm AWAY FROM THE FOOTING OVER A DISTANCE OF 1m AND SHAPED TO PREVENT PONDING OF WATER. ANY PERIMETER PAVING SHALL ALSO BE SUITABLY SLOPED AWAY FROM THE FOOTINGS.
3. WHERE PERIMETER PAVING SLABS ARE CONSTRUCTED THEY MUST HAVE A 10mm LAYER OF ABLEFLEX OR SIMILAR BETWEEN THE PAVING SLAB AND THE FOOTING/WALL. AN AGRICULTURAL DRAIN OR SIMILAR MUST BE INSTALLED AT THE OUTER EDGE OF THE PERIMETER OF THE PAVING SLAB AS PER DETAIL AND IS TO BE DIVERTED TO THE LEGAL POINT OF DISCHARGE.
4. WATERING AND GARDEN BEDS ARE NOT PERMITTED ADJACENT TO BUILDING(S) AND ADJACENT TO PERIMETER PAVING SLABS. PLUMBING AND DRAINAGE SUBSURFACE DRAINS SHALL NOT BE USED WITHIN 1.5m OF THE BUILDING UNLESS DESIGNED BY A QUALIFIED ENGINEER FAMILIAR WITH THIS TYPE OF DESIGN.
5. BULK EXCAVATIONS INCORPORATING CUT AND FILL MUST HAVE AN AGRICULTURAL DRAIN (OR SWALE DRAIN) AT THE BASE OF THE CUT, INTO THE NATURAL CLAY AND ALL SURFACE AND SUB-SURFACE WATER IS TO BE DIVERTED FROM THE FOOTING TO THE LEGAL POINT OF DISCHARGE. PROVISION OF A DISH DRAIN TO THE TOP OF ALL BATTERS WILL PREVENT THE OCCURRENCE OF SCOURING ON THE BATTER FACES.
6. THE BASE OF TRENCHES SHALL BE SLOPED AWAY FROM THE BUILDING. TRENCHES SHALL BE BACKFILLED WITH CLAY IN THE TOP 300mm WITHIN 1.5m OF THE BUILDING. COMPACTED CLAY SHOULD BE USED FOR BACKFILLING WHERE PIPES PASS BENEATH THE FOOTING SYSTEM THE TRENCHES WILL BE BACKFILLED FULL DEPTH WITH CLAY OR CONCRETE TO RESTRICT THE INGRESS OF WATER BENEATH THE FOOTING SYSTEM.
7. PENETRATIONS OF THE EDGE BEAMS OF RAFT OR PERIMETER STRIP FOOTINGS SHALL BE AVOIDED WHERE PRACTICABLE, BUT WHERE NECESSARY SHALL BE DETAILED TO ALLOW FOR MOVEMENT.
8. DRAINS ATTACHED TO/OR EMERGING FROM UNDERNEATH THE BUILDING SHALL INCORPORATE FLEXIBLE JOINTS IMMEDIATELY OUTSIDE THE FOOTING AND COMMENCING WITHIN 1m OF THE BUILDING PERIMETER TO ACCOMMODATE A TOTAL RANGE OF DIFFERENTIAL MOVEMENT IN ANY DIRECTION EQUAL TO THE ESTIMATED CHARACTERISTIC SURFACE MOVEMENT OF THE SITE (y_s). IN THE ABSENCE OF A SPECIFIC DESIGN GUIDANCE THE FITTINGS OR OTHER DEVICES THAT ARE PROVIDED TO ALLOW FOR MOVEMENT SHALL BE SET AT THE MID POSITION OF THEIR RANGE OF POSSIBLE MOVEMENT AT THE TIME OF INSTALLATION, SO AS TO ALLOW FOR MOVEMENT EQUAL TO $0.5y_s$ IN ANY DIRECTION FROM THE INITIAL SETTING. THIS REQUIREMENT APPLIES TO ALL STORMWATER, SEWER DRAINS AND DISCHARGE PIPES.

NOTE:
THE DRAINAGE DESIGN AS INDICATED ON THESE DRAWINGS RELATES ONLY TO UNPAVED AREAS AROUND THE PERIMETER OF THE BUILDING. IF PAVING OR LANDSCAPING IS TO BE INSTALLED AROUND THE PERIMETER AT A LATER DATE THEN THE DRAINAGE REQUIREMENTS NEED TO BE REVIEWED & UPGRADED TO ENSURE THAT ALL RUNOFF IS COLLECTED & DISPERSED ACCORDINGLY. PROPERTY OWNERS, CURRENT & FUTURE SHOULD BE MADE AWARE OF THIS REQUIREMENT.

NOTE:
IF ADDITIONAL PAVING OR LANDSCAPING IS TO BE INSTALLED AROUND THE PERIMETER AFTER HANDOVER THEN THE DRAINAGE REQUIREMENTS NEED TO BE REVIEWED & UPGRADED TO ENSURE THAT ALL RUNOFF IS COLLECTED & DISPERSED ACCORDINGLY. PROPERTY OWNERS, CURRENT & FUTURE SHOULD BE MADE AWARE OF THIS REQUIREMENT.

GENERAL NOTES:

1. ALL WORKS TO BE CARRIED OUT IN ACCORDANCE WITH THE LOCAL CITY COUNCIL STANDARD SPECIFICATIONS AND UNDER THE SUPERVISION OF THE CITY ENGINEER OR THEIR REPRESENTATIVES.
2. CONTRACTORS ARE TO VERIFY ALL EXISTING INFORMATION BEFORE COMMENCEMENT OF ANY WORK IN PARTICULAR THE LEGAL POINT OF DISCHARGE. IF ANY DISCREPANCY EXISTS BETWEEN THE ACTUAL LEGAL POINT OF DISCHARGE ON SITE, THIS PLAN AND/OR INFORMATION SUPPLIED BY COUNCIL, THEN THIS OFFICE SHOULD BE CONTACTED FOR FURTHER ADVICE. IT IS EMPHASISED THAT THE LEGAL POINT OF DISCHARGE IS TO BE CONFIRMED PRIOR TO ANY WORKS, INCLUDING FOUNDATIONS, SITE EXCAVATIONS AND RUNNING OF OTHER SERVICES, ETC.
3. OBTAIN ALL BUILDING SET-OUT DETAILS AND DIMENSIONS FROM ARCHITECTURAL DRAWINGS.
4. IT IS THE CONTRACTOR'S / CLIENT'S RESPONSIBILITY TO CHECK THE LOCATION AND DEPTHS OF ALL EXISTING UNDERGROUND SERVICES WHETHER SHOWN ON THIS DRAWING OR NOT, PRIOR TO THE COMMENCEMENT OF EXCAVATIONS.
5. LEVELS AND CONTOURS SHOWN ARE TO AN ARBITRARY DATUM.
6. EXISTING WORKS DAMAGED DURING CONSTRUCTION ARE TO BE REINSTATED TO ORIGINAL CONDITION BY CONTRACTOR TO THE SATISFACTION OF THE OWNER.
7. BACKFILL TRENCHES BENEATH PAVEMENT WITH F.C.R.
8. SUBGRADE TO BE STRIPPED OF TOP SOIL AND THOROUGHLY COMPACTED BEFORE PLACING PAVEMENT. PAVEMENT TO BE 100mm THICK CONCRETE WITH 1 LAYER OF SL72 MESH PLACED WITH 30mm TOP COVER ON 50mm COMPACTED DEPTH OF CRUSHED ROCK BEDDING.
9. 90mm DIA. MINIMUM AGGI DRAINS TO BE PLACED AT THE BASE OF ALL CUT BATTERS AND RETAINING WALLS, AND CONNECTED TO MAIN STORMWATER PIPES VIA 300mm DIA. SILT PITS, UNLESS NOTED OTHERWISE.
10. ALL WAFFLE EDGE & INTERNAL BEAMS ARE TO BE EXCAVATED DOWN 100mm MIN. BELOW THE 45° LINE OF INFLUENCE TO ANY NEW OR EXISTING ADJACENT SERVICE TRENCHES BUT ALSO INTO THE RECOMMENDED FOUNDING MATERIAL AS NOTED IN THE SOIL REPORT. THESE EXCAVATIONS MAY BE FILLED WITH BLINDING CONCERETE TO UNDERSIDE OF THE FOOTINGS OR INTERNAL BEAMS.

SITE CLASSIFICATION:
CLASS P (MOD. M)
CHARACTERISTIC SURFACE MOVEMENT (Ys):
20mm to 40mm

Drawn
SHAUN .F
Design
MANA .M
Date
13/09/21
Scale ---

Drainage Notes

Sheet No. **17**

Job No: **2214421/2**
SIMONDS HOMES EAST MELBOURNE
LOT 31, OSKAR COURT, PAKENHAM

ACN 162 735 917
1st FLOOR, 25-27 RAILWAY RD,
BLACKBURN, VIC. 3130
E: contact@clickeng.com.au
P: (03) 9894 4222

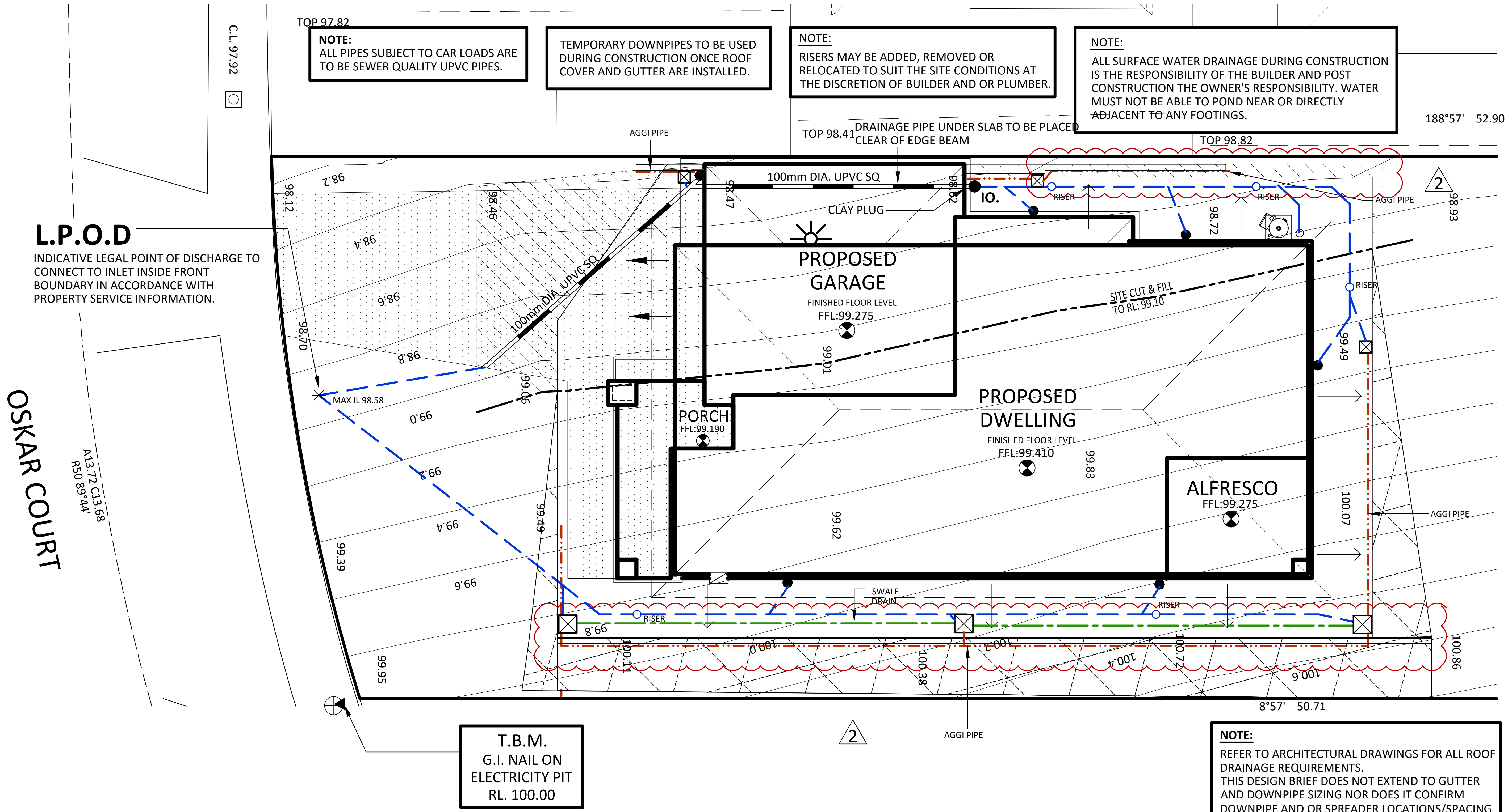
CLICK ENGINEERING

NOTE:
ALL PIPES SUBJECT TO CAR LOADS ARE TO BE SEWER QUALITY UPVC PIPES.

TEMPORARY DOWNPIPES TO BE USED DURING CONSTRUCTION ONCE ROOF COVER AND GUTTER ARE INSTALLED.

NOTE:
RISERS MAY BE ADDED, REMOVED OR RELOCATED TO SUIT THE SITE CONDITIONS AT THE DISCRETION OF BUILDER AND OR PLUMBER.

NOTE:
ALL SURFACE WATER DRAINAGE DURING CONSTRUCTION IS THE RESPONSIBILITY OF THE BUILDER AND POST CONSTRUCTION THE OWNER'S RESPONSIBILITY. WATER MUST NOT BE ABLE TO POND NEAR OR DIRECTLY ADJACENT TO ANY FOOTINGS.



NOTE:
REFER TO ARCHITECTURAL DRAWINGS FOR ALL ROOF DRAINAGE REQUIREMENTS.
THIS DESIGN BRIEF DOES NOT EXTEND TO GUTTER AND DOWNPIPE SIZING NOR DOES IT CONFIRM DOWNPIPE AND OR SPREADER LOCATIONS/SPACING AS DEPICTED ON THE ARCHITECTURAL DRAWINGS.
ALL THESE ELEMENTS MUST BE IN ACCORDANCE WITH AS3500.3 AND BE CONFIRMED BY A LICENSED PLUMBER.

LEGEND:-

- RISER — 100mm NOMINAL DIAMETER PVC PIPE 1:100 MIN. FALL UNLESS OTHERWISE NOTED
- 90mm NOMINAL DIAMETER AGGI PIPE 1 : 100 MIN. FALL
- SWALE DRAIN 1 IN 100 MIN. FALL
- INDICATIVE LOCATION OF CUT FILL LINE
- 300Ø SILT PIT
- IO. INSPECTION OPENING
- SPOT LEVEL
- 300 x 300 SILT PIT
- 450 x 450 SILT PIT
- TRENCH GRATE CONNECTED TO STORMWATER
- DIRECTION OF FALL, AWAY FROM FOOTINGS REFER TO DRAINAGE REQUIREMENTS ON SHEET 1
- DOWNPIPE LOCATIONS SHOWN ARE INDICATIVE ONLY AND AND CAN BE MOVED TO SUIT, PROVIDED THAT MINIMUM NUMBER SHOWN ARE PROVIDED.
- DOWNPIPE WITH SPREADER.
- DIRECTIONAL FALL OF PAVING.
- UPVC OR PVC PIPE USED AS APPROVED BY THE CITY ENGINEER. UPVC TO BE SEWER QUALITY.
- JUNCTION PITS AS SPECIFIED COMPLETE WITH CONCRETE COVER AND STEPIRONS .
- GRATING PITS AS SPECIFIED, COMPLETE WITH MEDIUM DUTY GRATE AND STEPIRONS.

NORTH

FILL NOTE:-
WHERE ADDITIONAL FILL IS PLACED OVER NATURAL SURFACE, THE NATURAL SURFACE IS TO FIRST BE SLOPED AWAY FROM BUILDING FOOTPRINT.

CLASS P (MOD. M)

Drawn
SHAUN .F
Design
MANA .M
Date
13/09/21
Scale 1:100

Roof And Site Drainage

Sheet No. **18**

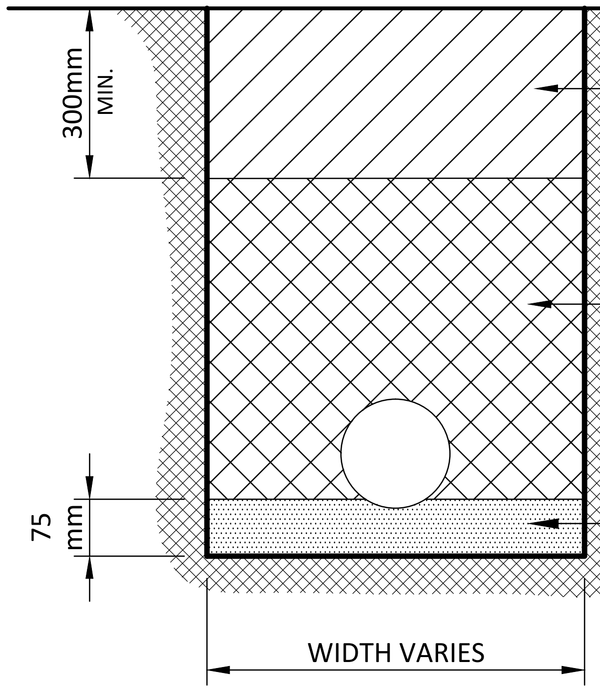
Job No: **2214421/2**

SIMONDS HOMES EAST MELBOURNE

LOT 31, OSKAR COURT, PAKENHAM

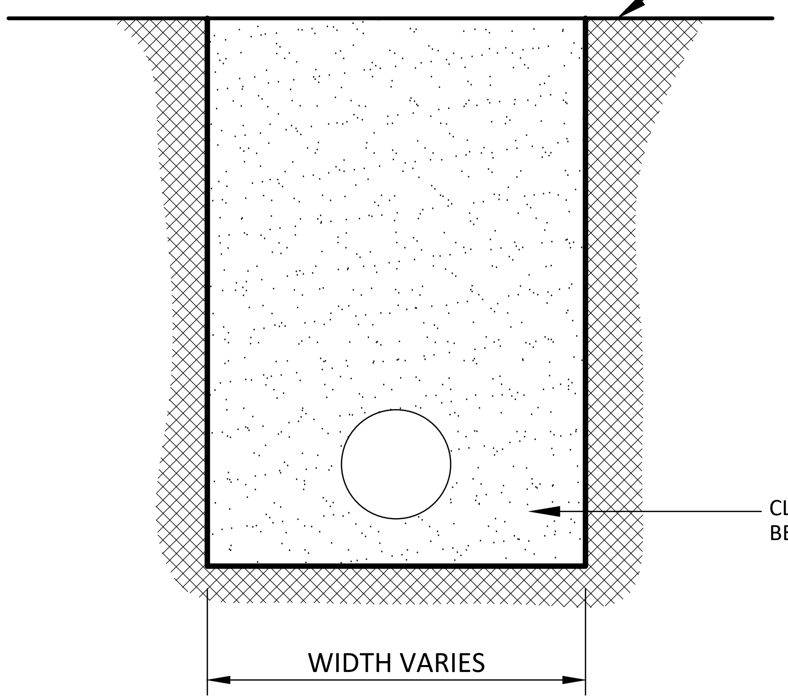
ACN 162 735 917
1st FLOOR, 25-27 RAILWAY RD,
BLACKBURN, VIC. 3130
E: contact@clickeng.com.au
P: (03) 9894 4222

CLICK ENGINEERING



TYPICAL PIPE TRENCH DETAIL N.T.S.

WHERE PIPES PASS UNDER THE FOOTING SYSTEM, THE TRENCH SHALL BE BACKFILLED FULL DEPTH WITH CLAY OR CONCRETE TO RESTRICT THE INGRESS OF WATER BENEATH THE FOOTING SYSTEM.



TYPICAL CLAY PLUG DETAIL N.T.S.

WHERE PIPE PASSES UNDER THE FOOTING SYSTEM

DRAINAGE SYSTEM

ALL SURFACE WATER DRAINAGE DURING CONSTRUCTION IS THE RESPONSIBILITY OF THE BUILDER. WATER MUST NOT BE ABLE TO POND NEAR OR DIRECTLY ADJACENT TO ANY FOOTINGS.
IN ADDITION TO THE DOCUMENTED DESIGN, AT THE DISCRETION OF THE BUILDER, WHERE A SITE CUT LESS THAN OR EQUAL TO 400mm OCCURS (MEASURED FROM THE WAFFLE CUT LEVEL), SILT PITS AND AGRICULTURAL DRAINS MAY NEED TO BE INSTALLED TO PREVENT PONDING OF SURFACE WATER AS REQUIRED.
IT IS ALSO THE RESPONSIBILITY OF THE BUILDER TO ENSURE NO RUNOFF IS ALLOWED TO ESCAPE THE SITE ONTO ADJOINING SITES OR PUBLIC AREAS.
PROVISIONS SUCH AS PHYSICAL BARRIERS AND OR AGRICULTURAL DRAINS/SILT PITS MAY BE REQUIRED AGAIN AT THE BUILDERS DISCRETION.
POST CONSTRUCTION IT IS THE OWNER'S RESPONSIBILITY TO MAINTAIN THE DRAINAGE SYSTEM UNTIL A LONG TERM LANDSCAPING SOLUTION IS PROVIDED.

FUTURE LANDSCAPING NOTE:

AFTER HANDOVER THE OWNER IS TO MAINTAIN A MINIMUM HEIGHT OF 150mm FROM THE FINISHED GROUND LEVEL, LANDSCAPING OR PAVING LEVEL TO THE DAMP PROOF COURSE. WHERE ADJOINING PAVED AREAS SLOPE AWAY FROM THE BUILDING, THESE HEIGHTS MAY BE REDUCED TO 75mm

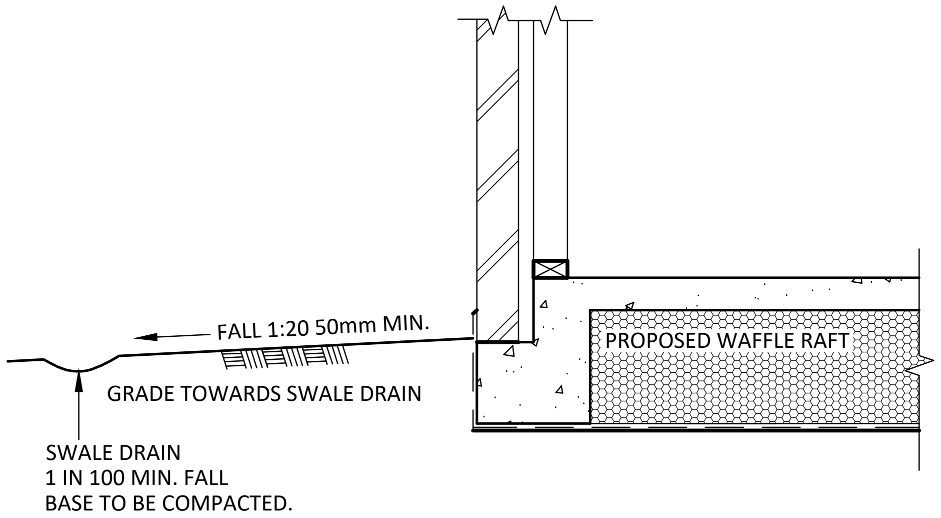
NOTE:
THE SOIL IS TO BE GRADED TO FALL 1:20 MINIMUM AWAY FROM THE FOOTING AND SHAPED TO PREVENT PONDING OF WATER. ANY PERIMETER PAVING SHALL ALSO BE SUITABLY SLOPED AWAY FROM THE FOOTINGS.

CLASS P (MOD. M)

Drawn SHAUN .F **Design** MANA .M **Date** 13/09/21 Scale ---	Drainage Details 1	Sheet No. **19** Job No: **2214421/2** SIMONDS HOMES EAST MELBOURNE **LOT 31, OSKAR COURT, PAKENHAM**

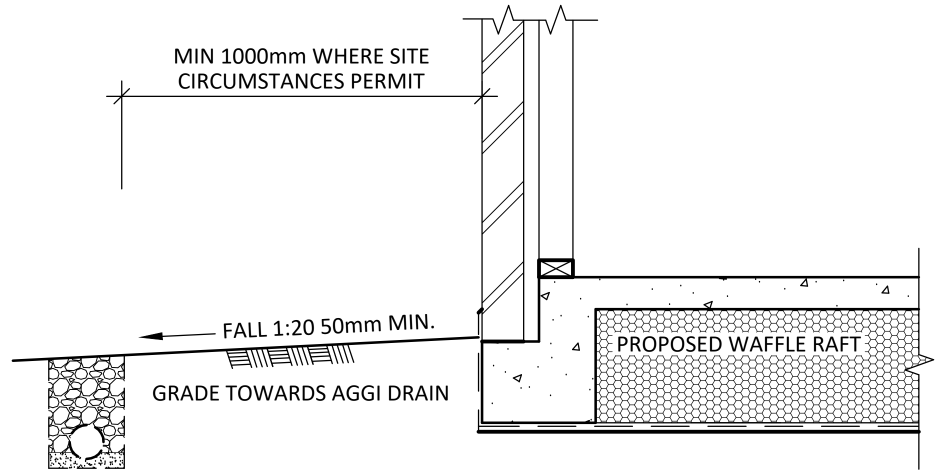
ACN 162 735 917
1st FLOOR, 25-27 RAILWAY RD,
BLACKBURN, VIC. 3130
E: contact@clickeng.com.au
P: (03) 9894 4222

CLICK ENGINEERING

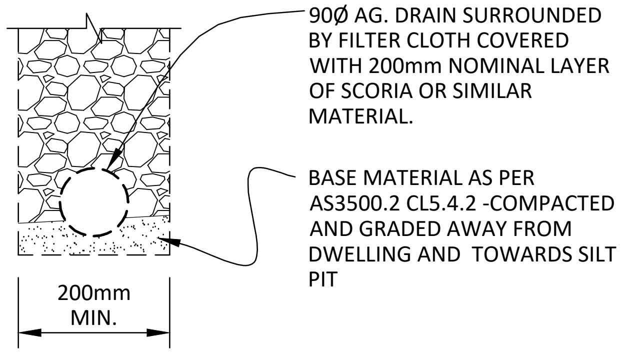


TYPICAL SWALE DRAIN. N.T.S.

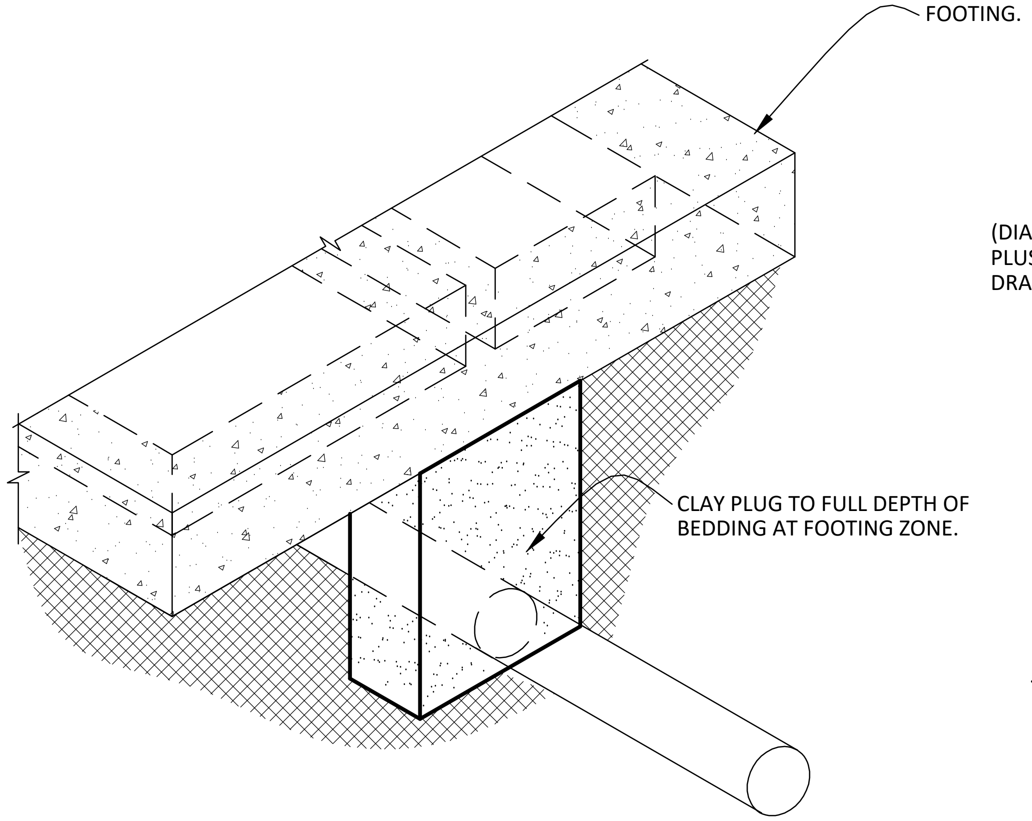
AS REQUIRED

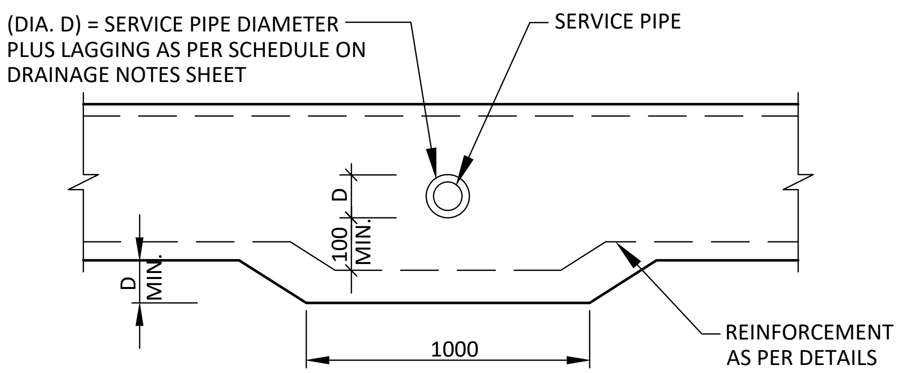


TYPICAL DRAINAGE. N.T.S.



TYPICAL AGGI. DRAIN





SERVICE PIPE LAGGING DETAIL

SERVICE EXIT DETAIL N.T.S.

CLASS P (MOD. M)

Drawn
SHAUN .F
Design
MANA .M
Date
13/09/21
Scale ---

Drainage Details 2

Sheet No. **20**

Job No: **2214421/2**

SIMONDS HOMES EAST MELBOURNE

LOT 31, OSKAR COURT, PAKENHAM

ACN 162 735 917
1st FLOOR, 25-27 RAILWAY RD,
BLACKBURN, VIC. 3130
E: contact@clickeng.com.au
P: (03) 9894 4222

CLICK ENGINEERING